

Earlier Split-Zero constructions in the RH programme: exact cohomology, boundary, source and metric receivers

Split-Zero research programme

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This edition gives complete receiving proofs from earlier Split-Zero manuscripts into the programme's original analytic and cohomological objects. It includes the original source coordinates, complete primary units, actual supported-cohomology and semimodule maps, arithmetic boundary morphisms, inverse-source singularities, holonomy sources, and finite metric and complex-response estimates. Human mathematical sources are cited at use. The accompanying source register distinguishes complete manuscript readings from selected complete sections; the corpus reading is still ongoing, and no exhaustive coverage is claimed.

The heat proof HM1–30 is reproduced from its sealed public edition for context and its exact dependencies; the missing typesetting backslash in the first order relation of HM19 is repaired here. The original edition is preserved. HD1–5 and HR1–11 carry its new receivers back into the actual finite source limits and projected currents. The original arithmetic allocation coefficients and individual signs remain to be evaluated; none are assigned from totals or dimensions.

Earlier corpus constructions and their exact RH receivers

This mathematical intake recovers several distinct constructions from earlier manuscripts and proves their maps into the theta-source and full-multiplicity zero objects used in the RH programme. It does not evaluate the currently unresolved seven absolute allocations or projected signs. All source versions and precise proof locators are recorded in the accompanying `SOURCE_RECEIVERS.md`. The statements below are proved here rather than inferred from names or similarities of constructions.

1 The original theta source from the earlier circle heat construction

Retain the physical variable $x > 0$, $D = -x\partial_x$, and

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad f_r = D^r \Theta\phi_*.$$

The source space is

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad \mathcal{H} = L^2(\mathbb{R}_{>0}, dx).$$

These are the original conventions in (TC1)–(TC6) of the theta-chain text. The older corrected spline manuscript uses the circle coordinate $z \in \mathbb{R}/\mathbb{Z}$,

$$h_0(z) = \mathbf{1}_{[0,1/2)}(z) - \mathbf{1}_{[1/2,1)}(z), \quad N = -\frac{1}{4\pi^2} \partial_z^2, \quad u_0(z, t) = e^{-tN} h_0(z).$$

Set $C(t) = \partial_z u_0(0, t)$ for $t > 0$.

Theorem 1.1 (Exact dyadic source and its full tail). *For each integer $J \geq 0$ and $r \geq 0$, put $s_J = 2^{J+1}$ and*

$$f_r^{[J]}(x) = \frac{1}{4} D^{r+1} (D - 1) \sum_{j=0}^J C(\pi 4^j x^2).$$

Then, with no rescaling of x in the receiving theta complex,

$$C(t) = 4 \sum_{\substack{k \in \mathbb{Z} \\ k \text{ odd}}} e^{-tk^2}, \tag{1}$$

$$f_r(x) - f_r^{[J]}(x) = f_r(s_J x), \tag{2}$$

$$\|f_r - f_r^{[J]}\|_{L^2(dx)} = s_J^{-1/2} \|f_r\|_{L^2(dx)}. \tag{3}$$

Moreover $f_r^{[J]} = \Theta(\phi_r - \phi_r(s_J \cdot))$ with $\phi_r = D^r \phi_ \in V$. Thus every truncation lies in the original theta image, and its complete Mellin transform is*

$$\mathcal{M}f_r^{[J]}(w) = (1 - s_J^{-w}) w^r 2\xi(w). \tag{D1}$$

Proof. For $k \neq 0$, direct integration gives $\hat{h}_0(k) = (1 - (-1)^k)/(\pi i k)$ and $\hat{h}_0(0) = 0$. The heat Fourier series and all its spatial derivatives converge absolutely for $t > 0$, giving (1). With $\vartheta(v) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 v}$, partitioning the integers into odd and even ones gives

$$\frac{1}{4} C(\pi 4^j x^2) = \vartheta(4^j x^2) - \vartheta(4^{j+1} x^2).$$

The finite sum therefore equals $\vartheta(x^2) - \vartheta(s_J^2 x^2)$ exactly. Also $D(D - 1)e^{-\pi x^2} = \phi_*(x)$; termwise differentiation of the Gaussian series on every compact subset of $(0, \infty)$ gives $D^{r+1}(D - 1)(\vartheta(x^2) -$

1) = $f_r(x)$. Since D commutes with $T_s F(x) = F(sx)$, this proves (2). The Gaussian source and its D -iterates have the two required zero moments. Poisson summation gives rapid decay of their theta sums at zero as well as infinity, so they belong to $\mathcal{H}$. Changing variables $y = s_J x$ proves (3).

Dilation preserves evenness, the value zero at zero and the zero integral of V . The sum defining Θ commutes with dilation, proving the displayed source identity. For $\Re w > 1$, termwise Mellin integration and two integrations by parts give

$$\mathcal{M}f_0(w) = w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w) = 2\xi(w).$$

Both sides extend to entire functions by the rapid decay just proved; the convention for ξ agrees with Apostol's formula recorded in [9]. Mellin transformation of D and T_s multiplies by w and s^{-w} respectively. This proves (D1). $\square$

Proposition 1.2 (Complete Gram receiver). *Let E be any one of the original finite packet coordinate spaces, $F_L : \mathbb{C}^{L+1} \rightarrow \mathcal{H}$, $F_L e_r = f_r$, and $C_E : E \rightarrow \mathbb{C}^{L+1}$ any fixed original coefficient map. Set $P = F_L C_E$, $P_J = (1 - T_{s_J})P$, and retain any original $R_{\text{ref}} : E \rightarrow \mathcal{H}$. For $R = R_{\text{ref}} + P$ and $R_J = R_{\text{ref}} + P_J$ one has*

$$\begin{aligned} \Delta_J &= T_{s_J} P, & \|\Delta_J\| &= s_J^{-1/2} \|P\|, \\ R_J^* R_J - R^* R &= -R^* \Delta_J - \Delta_J^* R + \Delta_J^* \Delta_J, \end{aligned} \tag{D2}$$

$$\|R_J^* R_J - R^* R\| \leq 2s_J^{-1/2} \|R\| \|P\| + s_J^{-1} \|P\|^2. \tag{D3}$$

The same exact dilation norm holds for the Hilbert–Schmidt norm. If $q : \mathcal{B} \rightarrow \mathcal{B}/\Theta V$ is the original quotient, then $qR_J = qR$. All their full Mellin jets at actual zeros of ξ agree.

Proof. $T_s^* T_s = s^{-1} I$ on $L^2(dx)$, which proves both norm equalities. Expand $(R - \Delta_J)^*(R - \Delta_J)$ without discarding either cross term to obtain (D2); the triangle and submultiplicative norm inequalities give (D3). Every column of Δ_J belongs to ΘV by the preceding theorem. Its Mellin transform is a polynomial times $s_J^{-w} 2\xi(w)$, so it vanishes to every original zero multiplicity. $\square$

The exact receiving map is $\phi_r \mapsto \phi_r - \phi_r(s_J \cdot)$ in V . It does not preserve the finite polynomial–Gaussian span $\text{span}(\phi_0, \dots, \phi_L)$: the second term has a different Gaussian scale. Consequently (D2) compares explicitly specified representatives of the same original theta class; it does not identify two finite-cutoff attained minima. The useful gain for the programme is a geometric heat realization of the full original source, with an exact error in the physical Hilbert norm and with all Mellin multiplicities preserved.

2 Full-multiplicity observations on the arithmetic-site approximation spaces

The earlier arithmetic-curve manuscript has the spaces

$$H_B = L^2((1, \infty), y^{-2} dy), \quad F_\alpha(y) = \alpha[y] - [\alpha y] = \{\alpha y\} - \alpha\{y\}, \quad 0 < \alpha \leq 1,$$

$$V_Q = \text{span}\{F_{a/Q} : 1 \leq a < Q\}, \quad x_Q = \text{dist}_{H_B}(1, V_Q)^2.$$

We retain this coordinate and measure; no identification with the theta Hilbert metric is implicit. This construction imports the human Nyman–Beurling and Báez-Duarte criterion, cited explicitly in [6, 7]. The new finite receiver below is proved independently of that equivalence.

Lemma 2.1 (Exact Mellin observation). *For $\Re w > 0$,*

$$\int_1^\infty F_\alpha(y) y^{-w-1} dy = \zeta(w) \frac{\alpha - \alpha^w}{w}. \tag{B1}$$

The apparent singularity at $w = 1$ on the right is removable.

Proof. F_α is bounded, so the left side is holomorphic on $\Re w > 0$. For $\Re w > 1$, write $\lfloor y \rfloor = \sum_{n \geq 1} \mathbf{1}_{[n, \infty)}(y)$ and $\lfloor \alpha y \rfloor = \sum_{n \geq 1} \mathbf{1}_{[n/\alpha, \infty)}(y)$. Absolute convergence justifies integration, yielding $\alpha \zeta(w)/w - \alpha^w \zeta(w)/w$. The numerator vanishes at 1, so analytic continuation proves (B1) on the stated half-plane. $\square$

Theorem 2.2 (Confluent Gram lower bound with every multiplicity). *Let Z_+ be a nonempty finite set of actual zeta zeros ρ with $\Re \rho > 1/2$, retaining their full multiplicities m_ρ . Use the original local coordinates $(w - \rho)^j$, $0 \leq j < m_\rho$, of $E_{Z_+} = \prod_{\rho \in Z_+} \mathbb{C}[w]/(w - \rho)^{m_\rho}$. Define*

$$(J_+ f)_{\rho, j} = \frac{1}{j!} \int_1^\infty f(y) (-\log y)^j y^{-\rho-1} dy, \quad b_{\rho, j} = \frac{(-1)^j}{\rho^{j+1}}.$$

Then $J_+ : H_B \rightarrow E_{Z_+}$ is a bounded surjection, $J_+ F_\alpha = 0$, $J_+ 1 = b$, and its positive definite Gram matrix is

$$G_{(\rho, j), (\sigma, k)} = \frac{(-1)^{j+k} (j+k)!}{j! k! (\rho + \bar{\sigma} - 1)^{j+k+1}}. \quad (\text{B2})$$

For every Q ,

$$x_Q \geq b^* G^{-1} b > 0. \quad (\text{B3})$$

In particular one simple observation at $\rho = \beta + i\gamma$ gives $x_Q \geq (2\beta - 1)/|\rho|^2$.

Proof. With inner products conjugate-linear in their first variable, the representer of the (ρ, j) observation is $k_{\rho, j}(y) = (-\log y)^j y^{1-\bar{\rho}}/j!$. Its squared norm is finite because $2\Re \rho - 1 > 0$. Thus $J_+^* v = \sum k_{\rho, j} v_{\rho, j}$. Setting $t = \log y$ and integrating $t^{j+k} e^{-at}$ gives (B2), with $a = \rho + \bar{\sigma} - 1$ of positive real part. The representers are independent: a linear relation is an exponential polynomial $\sum_\rho p_\rho(t) e^{(1-\bar{\rho})t} = 0$; successively applying the differential operators $(\partial_t - (1 - \bar{\sigma}))^{m_\sigma}$ for all but one exponent isolates a nonzero invertible operator on its polynomial coefficient. Every coefficient therefore vanishes. This proves $G > 0$ and $J_+ J_+^* = G$, hence surjectivity. Differentiating (B1) through order $m_\rho - 1$ proves $J_+ F_\alpha = 0$; all differentiation is justified by the logarithmic integrable bounds. Direct integration of 1 gives b .

The exact minimum-norm lift of b is $S_+ b = J_+^* G^{-1} b$. Every e with $J_+ e = b$ decomposes orthogonally as $S_+ b + (e - S_+ b)$, with the second summand in $\ker J_+$. Consequently $\|e\|^2 \geq b^* G^{-1} b$. Apply this to $e = 1 - v$, $v \in V_Q$, and take the minimum. Since the coordinate $b_{\rho, 0} = 1/\rho$ is nonzero, the lower bound is positive. For one value observation $G = (2\beta - 1)^{-1}$, giving the last formula. $\square$

Proposition 2.3 (The Beurling observation enters the original theta class). *For the same actual finite divisor Z_+ , use its original reference section R_{Z_+} of $(T2)$, with the unit $j_{h_{Z_+}}(h_{Z_+}/(2\xi))$. Then*

$$\mathcal{T}_+ = R_{Z_+} J_+ : H_B \longrightarrow \mathcal{B} \quad (\text{B4})$$

is continuous into the original test-function topology and satisfies

$$J_{\text{all}} \mathcal{T}_+ = \iota_+ J_+, \quad q \mathcal{T}_+ = \sigma_{Z_+} J_+, \quad \mathcal{T}_+ F_\alpha = 0, \quad q \mathcal{T}_+ 1 = \sigma_{Z_+} b \neq 0. \quad (\text{B5})$$

Here ι_+ is full-jet extension by zero, and all other maps are the original maps of the theta complex. Thus (B3) is the exact attained quotient metric of a second analytic presentation of these same finite theta classes.

Proof. J_+ is bounded and has finite-dimensional codomain. A linear map from that finite-dimensional space into $\mathcal{B}$ is continuous; concretely every defining seminorm of $R_{Z_+} v$ is bounded by a finite sum of the seminorms of its fixed columns times the coordinates of v . The identities follow from Theorem 2.2 and (T2), including the original complete local unit. Injectivity of σ_{Z_+} and $b \neq 0$ prove the final nonvanishing. The quotient norm on $H_B/\ker J_+$ is exactly $v^* G^{-1} v$ by the minimum-norm lift proved above. This assertion specifies that quotient metric; it does not equate it with any theta L^2 metric. $\square$

Proposition 2.4 (Exact arithmetic-site tower receiver). *Put $\Lambda_Q = \text{lcm}(1, \dots, Q)$ and $R_Q = \text{span}\{F_{1/n} : 2 \leq n \leq Q\}$. Then $R_Q \subset V_{\Lambda_Q}$ and, for $d_Q = \text{dist}(1, R_Q)$, $x_{\Lambda_Q} \leq d_Q^2$. For $Q_0 \geq 3$, $Q_{j+1} = \Lambda_{Q_j}$ is strictly increasing and every positive integer divides some Q_j . Every off-line actual zero supplies the fixed positive lower bound (B3) at all these levels, using its right-half-plane functional-equation partner when necessary.*

Proof. For $n \leq Q$, $1/n = (\Lambda_Q/n)/\Lambda_Q$, which gives the inclusion and distance inequality. Because Q and $Q-1$ are coprime, $\Lambda_Q \geq Q(Q-1) > Q$ for $Q \geq 3$. Hence the tower grows without bound and once $Q_j \geq n$, $n \mid Q_{j+1}$. If $\Re \rho < 1/2$, $1-\rho$ has real part greater than $1/2$ and the same full multiplicity, by the functional equation [9]. Apply the preceding theorem at that actual zero; no hypothetical replacement spectrum is used. $\square$

For clarity, the relation to the usual human criterion is exact. Under the isometry $Uf(x) = f(1/x)$ from H_B to $L^2(0, 1)$, $UF_{1/n} = \rho_n - n^{-1}\rho_1$, where $\rho_n(x) = \{1/(nx)\}$. Extend this difference by zero to $(1, \infty)$. If $g = \sum c_n \rho_n$ on $(0, \infty)$ and $A = \sum c_n/n$, then $g(x) = A/x$ for $x > 1$, and $g - A\rho_1 = \sum c_n(\rho_n - n^{-1}\rho_1)$ is supported in $(0, 1)$. For $\chi = \mathbf{1}_{(0,1)}$, $|A| = \|g - \chi\|_{L^2(1, \infty)}$, so

$$\|g - A\rho_1 - \chi\|_{L^2(0, \infty)} \leq (1 + \|\rho_1\|_2)\|g - \chi\|_2.$$

Thus the compensated spaces have exactly the same closure criterion as the integer spaces in Báez-Duarte's theorem. Together with the finite divisibility inclusion above this proves that decay $x_{Q_j} \rightarrow 0$ on the cofinal tower is precisely this established RH criterion. No decay estimate has been assumed or proved here. What is gained is an exact finite, full-multiplicity observation and its computable positive obstruction at every arithmetic-site stage. In the arithmetic-curve construction the framed groups $(1/Q)\mathbb{Z}$ and monoids $(1/Q)\mathbb{N}$ carry these same stages; the transition $Q \mid M$ becomes the monoid Frobenius $n \mapsto (M/Q)n$ in their framed integral coordinates.

3 The old compactified jet tail receives the actual theta quotient

Retain the original spaces

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^r F(x)| < \infty \text{ for every } b \in \mathbb{Z}, r \geq 0\}, \quad Q = \mathcal{B}/\Theta V.$$

Let Z_∞ be the discrete set of all actual nontrivial zeros, $E_\rho = \mathbb{C}[w]/(w - \rho)^{m_\rho}$, and

$$E_{\text{fin}} = \bigoplus_{\rho \in Z_\infty} E_\rho, \quad E_{\text{all}} = \prod_{\rho \in Z_\infty} E_\rho, \quad T = E_{\text{all}}/E_{\text{fin}}.$$

The earlier packet-geometry manuscript constructs exactly this last tail as the stalk at infinity of its compactified product-of-jets sheaf. The following receiver retains both the global tail and the analytic kernel invisible to every zero jet.

Theorem 3.1 (Global full-jet and tail receiver). *The map*

$$J_{\text{all}} : \mathcal{B} \longrightarrow E_{\text{all}}, \quad F \longmapsto (j_\rho \mathcal{M}F)_\rho$$

annihilates ΘV and descends to a $\mathbb{C}[w]$ -module map $\bar{J} : Q \rightarrow E_{\text{all}}$, with w acting by D on Q . The original finite reference sections induce a canonical injective $\mathbb{C}[w]$ -module map $\sigma : E_{\text{fin}} \rightarrow Q$ satisfying $\bar{J}\sigma = \iota$, the finite-support inclusion. Writing $K = \ker \bar{J}$ and $T_{\text{arith}} = \text{im}(Q \xrightarrow{\bar{J}} E_{\text{all}} \rightarrow T)$, there is an exact sequence

$$0 \longrightarrow K \longrightarrow Q/\sigma(E_{\text{fin}}) \longrightarrow T_{\text{arith}} \longrightarrow 0, \quad T_{\text{arith}} \hookrightarrow T. \quad (\text{T1})$$

The space T_{arith} is nonzero.

Proof. Every Mellin transform of $F \in \mathcal{B}$ is entire and can be differentiated under its integral, by the bounds defining $\mathcal{B}$ at both endpoints. For $\phi \in V$, the original Mellin identity is $\mathcal{M}\Theta\phi(w) = 2\zeta(w)\mathcal{M}\phi(w)$ for $\Re w > 1$. The function $\mathcal{M}\phi$ is holomorphic for $\Re w > -2$, since ϕ is even and vanishes to order at least two at zero. It follows by continuation that the displayed product vanishes to every full nontrivial-zero multiplicity. Thus $J_{\text{all}}\Theta = 0$. Integration by parts gives $\mathcal{M}DF(w) = w\mathcal{M}F(w)$ with vanishing endpoint terms. This proves the module assertion.

For a finite subset Z , let $h_Z(w) = \prod_{\rho \in Z}(w - \rho)^{m_\rho}$, $d_Z = \deg h_Z$, and $g(w) = 2\xi(w)$. If $u \in E_Z = \mathbb{C}[w]/h_Z$, let $r_Z(u)$ be the unique polynomial of degree $< d_Z$ representing $j_{h_Z}(h_Z/g)u$. The original reference section $R_Z : E_Z \rightarrow \mathcal{B}$ has

$$\mathcal{M}R_Z u(w) = \frac{g(w)r_Z(u)(w)}{h_Z(w)}, \quad qDR_Z = qR_Z M_w. \quad (\text{T2})$$

For completeness, the division operator giving (T2) is $S_\rho F(x) = x^{-\rho} \int_x^\infty F(y)y^{\rho-1}dy$ when $\mathcal{M}F(\rho) = 0$. It satisfies $(D - \rho)S_\rho F = F$. The zero moment rewrites its integral as minus the integral from zero to x , proving rapid decrease at zero; the original formula proves it at infinity. Iterating through the full multiplicities gives the division by h_Z in (T2). The source before division is $\Theta(r_Z(u)(D)\phi_*)$, whose Mellin transform is $gr_Z(u)$. Finally the remainder of multiplication of $r_Z(u)$ by w is $r_Z(M_w u)$ modulo h_Z , with quotient its original highest coefficient. It follows that $DR_Z - R_Z M_w$ is a scalar multiple of f_0 , which lies in ΘV . This proves the second assertion in (T2).

At zeros in Z , (T2) has exactly the jets u , by the unit h_Z/g ; at every other zero it vanishes to the full multiplicity. Hence $J_{\text{all}}R_Z u$ is the extension of u by zero. If $Z \subset Z'$ and u' is extension by zero, then

$$r_{Z'}(u') = r_Z(u) \frac{h_{Z'}}{h_Z}.$$

Indeed the right side has degree $< d_{Z'}$ and satisfies all defining jets, including the new zero jets. Substitution in (T2) gives equal Mellin transforms and thus equal reference sections (Mellin uniqueness on any vertical line follows from Fourier uniqueness in $\log x$). The qR_Z therefore assemble into σ . Its full-jet left inverse proves injectivity and $K \cap \sigma(E_{\text{fin}}) = 0$.

An element of Q maps to zero in T exactly when its full-jet family has finite support. Subtracting the corresponding σ section puts it in K . Thus the kernel before quotient is $K + \sigma(E_{\text{fin}})$, and after quotient it is canonically K . This proves every assertion of exactness in (T1).

To see nonzero arithmetic tails directly, choose a nonzero nonnegative $\chi \in C_c^\infty((1, 2))$. Its Mellin transform $H(w)$ is entire and is not identically zero, since $H(a) > 0$ for real a . For every actual zero ρ , the function $a \mapsto H(a + \rho)$ has only a discrete set of real zeros. The union of these exceptional sets over the countable Z_∞ is countable. Choose a real a outside it and put $F(x) = x^a \chi(x) \in \mathcal{B}$. Then every value coordinate of $J_{\text{all}}F$ is nonzero. Since the nontrivial zero set is infinite (the classical zero-counting theorem), this family has infinite support and hence nonzero image in T . $\square$

Here is the sheaf-level target explicitly. For $j : \mathbb{C} \hookrightarrow \mathbb{P}^1(\mathbb{C})$ and $\mathcal{F}(U) = \prod_{\rho \in U \cap Z_\infty} E_\rho$, the old manuscript proves

$$0 \rightarrow j_! \mathcal{F} \rightarrow j_* \mathcal{F} \rightarrow i_{\infty*} T \rightarrow 0. \quad (\text{T3})$$

At a finite zero both stalks equal its full E_ρ ; away from zeros both vanish. At infinity the first stalk vanishes, and the second is the product modulo finite modification: a neighborhood excludes only finitely many points of the closed discrete support. This proves (T3) on stalks. Globally $\Gamma(j_! \mathcal{F}) = E_{\text{fin}}$ and $\Gamma(j_* \mathcal{F}) = E_{\text{all}}$, so (T1) is a receiver into the actual global boundary of (T3). Neither $K = 0$ nor $T_{\text{arith}} = T$ is asserted. This retains an analytic part and a global boundary that would both be lost by identifying the programme with a finite zero divisor.

4 Both endpoint states and the complete even-Schwartz source

The earlier spectral-packet paper invokes the trace construction of Connes and Consani. Their actual source range theorem, [10, Lemma 4.1], retains both evaluation at zero and integration. Here is the full receiver in the present programme's physical coordinate and with its factor two. Let $\mathcal{S}_{\text{ev}} = \mathcal{S}(\mathbb{R})^{\text{ev}}$, and choose a smooth η on $(0, \infty)$ equal to 1 near zero and to 0 for $x \geq 2$. Define the direct vector-space sum

$$\mathcal{B}_{01} = \mathcal{B} \oplus \mathbb{C}\eta \oplus \mathbb{C}x^{-1}\eta, \quad b(F + c_0\eta + c_1x^{-1}\eta) = (c_0, c_1).$$

Its definition is independent of η : two choices differ by an element of $\mathcal{B}$, as do their products with x^{-1} .

Theorem 4.1 (Exact two-endpoint receiver). *The original theta sum extends to $\Theta : \mathcal{S}_{\text{ev}} \rightarrow \mathcal{B}_{01}$, and the following rows and boundary map are exact:*

$$\begin{array}{ccccccc} 0 & \rightarrow & V & \rightarrow & \mathcal{S}_{\text{ev}} & \xrightarrow{a} & \mathbb{C}^2 & \rightarrow & 0 \\ & & \Theta \downarrow & & \Theta \downarrow & & \text{diag}(-1, 1) \downarrow & & \\ 0 & \rightarrow & \mathcal{B} & \rightarrow & \mathcal{B}_{01} & \xrightarrow{b} & \mathbb{C}^2 & \rightarrow & 0, \end{array} \quad a(\phi) = \left(\phi(0), \int_{\mathbb{R}} \phi \right). \quad (\text{E1})$$

Inclusion of the original target induces the canonical isomorphism

$$\mathcal{B}/\Theta V \xrightarrow{\sim} \mathcal{B}_{01}/\Theta \mathcal{S}_{\text{ev}}. \quad (\text{E2})$$

It intertwines the original $D = -x\partial_x$, and the action of D on each endpoint quotient $\mathbb{C}^2$ is $\text{diag}(0, 1)$. The meromorphic Mellin transform of $F \in \mathcal{B}_{01}$ has precisely the possible endpoint residues

$$\text{Res}_{w=0} \mathcal{M}F = c_0, \quad \text{Res}_{w=1} \mathcal{M}F = c_1. \quad (\text{E3})$$

Proof. Poisson summation in the Fourier convention $\hat{\phi}(t) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x t} dx$ gives

$$\Theta \phi(x) = x^{-1} \int_{\mathbb{R}} \phi - \phi(0) + x^{-1} \Theta \hat{\phi}(1/x).$$

The last term and all its D -derivatives decay more rapidly than every power at zero. The defining theta series gives the corresponding decay at infinity. This proves the middle arrow and the boundary matrix in (E1). The source boundary map is onto: the Gaussian $g = e^{-\pi x^2}$ and $x^2 g$ have boundary pairs $(1, 1)$ and $(0, 1/(2\pi))$. The remaining exactness statements follow from the definitions.

To prove (E2) without discarding a boundary component, take any $F \in \mathcal{B}_{01}$. Choose $\phi \in \mathcal{S}_{\text{ev}}$ satisfying $a(\phi) = \text{diag}(-1, 1)b(F)$. Then $F - \Theta\phi \in \mathcal{B}$, proving surjectivity. If $F \in \mathcal{B}$ equals $\Theta\phi$ for some $\phi \in \mathcal{S}_{\text{ev}}$, the invertible boundary matrix forces $a(\phi) = 0$, so $\phi \in V$. This proves injectivity and proves that a choice of boundary lift does not change the quotient map.

D commutes with Θ . Also $(D\phi)(0) = 0$ and $\int_{\mathbb{R}} D\phi = \int_{\mathbb{R}} \phi$ by integration by parts, while $D1 = 0$ and $Dx^{-1} = x^{-1}$. Derivatives of the cutoff contribute only elements of $\mathcal{B}$, proving the asserted quotient actions. Finally subtract $c_0 + c_1x^{-1}$ near zero inside the Mellin integral. The remainder is entire in w , and integration over a sufficiently small interval $(0, \epsilon)$ gives $c_0\epsilon^w/w + c_1\epsilon^{w-1}/(w-1)$. This proves (E3), retaining both signs. $\square$

Proposition 4.2 (Original Laplacian range, with explicit inverses). *The operator $D(D-1) : \mathcal{S}_{\text{ev}} \rightarrow V$ is a bijection. For $\psi \in V$, its inverse is constructed on $x > 0$ by*

$$g(x) = \int_x^\infty \frac{\psi(t)}{t} dt, \quad f(x) = x^{-1} \int_x^\infty g(t) dt, \quad (\text{E4})$$

followed by smooth even extension at zero. In particular $\Theta D(D-1)e^{-\pi x^2} = f_0$, with exactly the original source and its completed Mellin transform 2ξ .

Proof. Since $\psi(0) = 0$ and ψ is even, $\psi(t)/t$ has a smooth odd extension at zero. The first integral is smooth even there and rapidly decreasing at infinity, and $Dg = \psi$. Fubini gives $\int_0^\infty g(t)dt = \int_0^\infty \psi(t)dt = 0$. Consequently the second expression equals $-x^{-1} \int_0^x g(t)dt$, which is smooth and even at zero. Its expression as a tail integral proves Schwartz decay at infinity, and $(D-1)f = g$. Thus $D(D-1)f = \psi$. Conversely $D(D-1)f$ has value zero at zero and zero integral, so lies in V . The equation $D(D-1)f = 0$ on $(0, \infty)$ has general solution $c + d/x$. Smoothness at zero and decay at infinity force $c = d = 0$, proving uniqueness. $\square$

The precise unweighted/weighted trace dictionary is $E\phi(x) = \frac{1}{2}x^{1/2}\Theta\phi(x)$ and $H = x\partial_x = -D$; hence $H(1+H) = D(D-1)$ before conjugating by multiplication by $x^{1/2}$. This is an operator identity, not an identification of unrelated Hilbert metrics. The Connes–Consani spectral quotient uses the closure of the source range in its specified strong-Schwartz topology. For the original $\mathcal{B}$ topology there is always the exact map

$$0 \rightarrow \overline{\Theta V}/\Theta V \rightarrow Q \rightarrow \mathcal{B}/\overline{\Theta V} \rightarrow 0. \quad (\text{E5})$$

Every original finite Mellin jet is continuous in that topology and annihilates $\overline{\Theta V}$, so the finite sections (T2) survive this passage. No equality between algebraic and closed images is assumed. For the Gaussian, (E3) gives residues $(-1, 1)$ and the endpoint principal part $1/(w(w-1))$. Thus the earlier quotient/lift residue calculation and the current theta-source construction have the same exact two-endpoint complex, with their common boundary explicitly removed by $D(D-1)$.

5 The old local phase germ gives every entry of the original Taylor unit

The atomic-phase manuscript retains, at a zero ρ of exact multiplicity m_ρ , the complete holomorphic factorization

$$\xi(\rho + z) = \lambda_\rho z^{m_\rho} \exp\left(\sum_{n \geq 1} c_{\rho,n} z^n\right), \quad \lambda_\rho = \frac{\xi^{(m_\rho)}(\rho)}{m_\rho!}. \quad (\text{U0})$$

Its phase-walk interpretation is explicitly credited there to unpublished correspondence of Jacolm Tobley. The receiver here uses the full complex logarithmic germ (U0); it makes no claim that the absolute packet weight $|\lambda_\rho|$ determines this germ.

Theorem 5.1 (All entries of the original unit, without removing phases). *Retain a finite actual packet divisor $h_Z(w) = \prod_{\rho \in Z} (w - \rho)^{m_\rho}$ of degree d and $g = 2\xi$. For $\rho \in Z$, define*

$$H_\rho = \prod_{\sigma \in Z \setminus \{\rho\}} (\rho - \sigma)^{m_\sigma}, \quad S_{\rho,n} = \sum_{\sigma \neq \rho} \frac{m_\sigma}{(\rho - \sigma)^n}, \quad a_{\rho,n} = \frac{(-1)^{n-1}}{n} S_{\rho,n} - c_{\rho,n}.$$

The complete local jet of the original multiplier $\varepsilon_Z = j_{h_Z}(h_Z/g)$ has coefficients

$$\varepsilon_{\rho,0} = \frac{H_\rho}{2\lambda_\rho}, \quad \varepsilon_{\rho,r} = \frac{1}{r} \sum_{n=1}^r n a_{\rho,n} \varepsilon_{\rho,r-n} \quad (1 \leq r < m_\rho). \quad (\text{U1})$$

Let V_Z be the actual confluent evaluation matrix from the original increasing power basis to the divided Taylor coefficients, namely

$$(V_Z)_{(\rho,j),\ell} = \begin{cases} \binom{\ell}{j} \rho^{\ell-j}, & \ell \geq j, \\ 0, & \ell < j, \end{cases} \quad 0 \leq \ell < d.$$

Let T_ρ be the lower triangular matrix with $(T_\rho)_{rj} = \varepsilon_{\rho, r-j}$ for $r \geq j$ and zero otherwise. For the divided-jet input $u \in E_Z$ used in (T2), the exact jet-to-power-coefficient map is

$$\text{coeff}(r_Z(u)) = V_Z^{-1} \left(\bigoplus_{\rho \in Z} T_\rho \right) u.$$

If instead the input is the increasing-power coefficient vector a of the class whose jets are $u = V_Z a$, the exact multiplication matrix is

$$M_{\varepsilon_Z} = V_Z^{-1} \left(\bigoplus_{\rho \in Z} T_\rho \right) V_Z. \quad (\text{U2})$$

Thus the two displayed matrices have different specified input coordinates; the intervening morphism is precisely the confluent evaluation V_Z . All root differences, full multiplicities and complex phases occur in this original-coordinate formula. If Z is stable under the two functional-equation symmetries, then

$$\varepsilon_{1-\rho, r} = (-1)^{d+r} \varepsilon_{\rho, r}, \quad \varepsilon_{\bar{\rho}, r} = \overline{\varepsilon_{\rho, r}}. \quad (\text{U3})$$

Proof. Every factor $\rho - \sigma$ is nonzero. On a sufficiently small disc, the logarithm with value zero of $\prod_{\sigma \neq \rho} (1 + z/(\rho - \sigma))^{m_\sigma}$ has coefficient $(-1)^{n-1} S_{\rho, n}/n$. Dividing this product by the nonvanishing unit in (U0), and retaining the factor 2 in $g = 2\xi$, gives

$$\frac{h_Z(\rho + z)}{g(\rho + z)} = \frac{H_\rho}{2\lambda_\rho} \exp \left(\sum_{n \geq 1} a_{\rho, n} z^n \right).$$

Differentiate this convergent identity and compare coefficients of z^{r-1} to obtain (U1). Multiplication of truncated Taylor polynomials is the Cauchy convolution of their coefficients, precisely T_ρ . The map V_Z is injective: a polynomial of degree less than d with all the stated zero jets is divisible by the monic degree- d polynomial h_Z and hence vanishes. Equal finite dimensions give invertibility, proving (U2). Finally, for a symmetry-stable divisor, $h_Z(1 - w) = (-1)^d h_Z(w)$, $g(1 - w) = g(w)$, and both h_Z, g commute with conjugation. Apply these identities at $1 - \rho + z$ and $\bar{\rho} + z$ and compare Taylor coefficients to obtain (U3). $\square$

This receiver converts a previously separate phase manuscript into actual entries of the programme's canonical finite lift. In particular it is available before any metric or determinant calculation: the lifted source is exactly $\Theta(r_Z(u)(D)\phi_*)$, divided by $h_Z(D)$ as in (T2). The determinant of M_{ε_Z} equals $\prod_\rho \varepsilon_{\rho, 0}^{m_\rho}$, but the full source and its Gram cross terms retain the remaining coefficients in (U1). No estimate on that determinant is substituted for an estimate on the complete map (U2).

6 The old jet-ideal chain enters Deligne's nilpotent filtration

The earlier Boolean-jet and refined-divisor manuscripts prove that the ideals of $\mathbb{C}[t]/t^m$ form the intrinsic chain $I_j = (t^j)$, and give the adjunctions from this chain to the Boolean multiplicity cube. Here this chain is carried into the original finite theta class and its actual metric. The integer grading below is precisely the nilpotent grading of Deligne [12, (1.6.1), (1.6.7)]; it does not assign an arithmetic Frobenius weight to a zeta zero.

Retain $E_Z = \mathbb{C}[w]/h_Z$, its original increasing-power basis, and the confluent matrix V_Z of (U2). On its actual primary factor at ρ write $t = w - \rho$ and $N_\rho u = tu$. Define

$$M_r E_\rho = \text{span}\{t^j : 0 \leq j < m_\rho, m_\rho - 1 - 2j \leq r\}, \quad M_r E_Z = V_Z^{-1} \bigoplus_\rho M_r E_\rho. \quad (\text{N1})$$

In particular the exponent of the ideal in the first formula is exactly $\max(0, \min(m_\rho, \lceil (m_\rho - 1 - r)/2 \rceil))$. Let $N = V_Z^{-1}(\bigoplus N_\rho)V_Z$; this removes the semisimple part by its actual primary idempotents, without changing $A = M_w$.

Theorem 6.1 (Full multiplicity filtration and its original receiver). *The filtration (N1) satisfies*

$$NM_r \subset M_{r-2}, \quad N^r : \mathrm{Gr}_r^M E_Z \xrightarrow{\sim} \mathrm{Gr}_{-r}^M E_Z \quad (r \geq 0). \quad (\text{N2})$$

It is Deligne's canonical nilpotent filtration. It is preserved by the original Taylor-unit multiplication M_{ε_Z} , whose action on the grade belonging to (ρ, j) is the full scalar $\varepsilon_{\rho,0} = H_\rho/(2\lambda_\rho)$ from (U1). The injective original section $\sigma : E_Z \rightarrow Q$ of (T2) carries it to an A -stable filtration of $\sigma(E_Z)$. On that image the action is the restriction of the original D , and N is transported by the actual spectral idempotents of that restriction.

Proof. Multiplication by t lowers the displayed weight by two, proving the inclusion. If $r \geq 0$, the source grade at ρ is zero unless $r \leq m_\rho - 1$ and $r \equiv m_\rho - 1 \pmod{2}$. In the remaining case it is generated by $t^{(m_\rho-1-r)/2}$. Its image under N^r is $t^{(m_\rho-1+r)/2}$, the nonzero generator of the target grade. This proves every isomorphism with its coefficient equal to one in the original divided Taylor coordinates. Deligne's uniqueness theorem identifies this verified filtration with his canonical one.

For clarity, uniqueness on each actual cyclic primary can also be seen directly. An N_ρ -stable subspace is an ideal of $\mathbb{C}[t]/t^m$, so the weights a_j at which t^j enters any such filtration satisfy $a_{j+1} \leq a_j - 2$. The graded isomorphisms force the largest and smallest weights to be b and $-b$. The nonzero map N^b from the top grade implies $b \leq m - 1$; the $m - 1$ successive drops imply $b \geq m - 1$. Equality holds and every drop is exactly two, giving $a_j = m - 1 - 2j$. This proves uniqueness directly among the original primary-stable filtrations as well.

In each primary, multiplication by the unit $\sum \varepsilon_{\rho,j} t^j$ preserves every ideal (t^j) . Its inverse is again a unit, so preservation is equality. Modulo the next nonzero ideal step only its constant term survives. This proves the asserted full scalar, including its phase and root-difference factors. The identity $qDR_Z = qR_ZA$, proved in (T2), transports the entire filtration into the original quotient. Since the different primary eigenvalues are distinct, their idempotents are the actual Chinese-remainder polynomials in A ; applying those same polynomials in $D|_{\sigma(E_Z)}$ gives exactly the stated transported N . $\square$

There is an explicit raising operator on the associated graded, without introducing a replacement metric:

$$R_\rho t^j = j(m_\rho - j)t^{j-1}, \quad H_\rho^{\mathrm{gr}} t^j = (m_\rho - 1 - 2j)t^j. \quad (\text{N3})$$

The value is zero for $j = 0$. Direct substitution gives $[H^{\mathrm{gr}}, N] = -2N$, $[H^{\mathrm{gr}}, R] = 2R$, and $[R, N] = H^{\mathrm{gr}}$. For example the last diagonal coefficient is $(j+1)(m-j-1) - j(m-j) = m-1-2j$, including the two endpoint cases. Thus the old chain supplies the exact $\mathfrak{sl}_2$ grading used in Deligne (1.6); these formulas do not assert $R = N^*$ for the original arithmetic metric.

Proposition 6.2 (Every graded quotient retains its original metric). *For this metric assertion let h_{act} be any of the programme's actual monic finite root polynomials, retaining every root κ and its full multiplicity m_κ . Write $E_{\mathrm{act}} = \mathbb{C}[w]/h_{\mathrm{act}}$ and let V_{act} be its own confluent evaluation matrix, exactly as in (U2). Formula (N1), with these roots and multiplicities, defines its nilpotent filtration. Let G_N be its existing positive definite attained arithmetic metric in the original increasing-power coefficients. Thus this assertion applies directly to the full sum polynomial and G_{q-1} of CMO1; it does not identify the sum roots with zeros of ξ or apply (U1) to a polynomial that does not divide 2ξ . Order the pairs (κ, j) by their actual weights $m_\kappa - 1 - 2j$, using a fixed order inside ties, and let C be the matrix with columns $V_{\mathrm{act}}^{-1} e_{\kappa,j}$ in this order. Put $\mathcal{H} = C^* G_N C$. For a weight r write $<$ for all earlier grades and r for the current grade. The metric on the actual quotient M_r/M_{r-1} is exactly*

$$\mathcal{G}_r = \mathcal{H}_{rr} - \mathcal{H}_{r<} \mathcal{H}_{< r}^{-1} \mathcal{H}_{< r}. \quad (\text{N4})$$

For the first grade the subtracted term is zero. Moreover

$$\det G_N = |\det V_{\text{act}}|^2 \prod_r \det \mathcal{G}_r. \quad (\text{N5})$$

For the exact induced map $T_r = N^r : \text{Gr}_r \rightarrow \text{Gr}_{-r}$, its squared norm on a coefficient vector u is $(T_r u)^* \mathcal{G}_{-r} (T_r u)$, and its domain squared norm is $u^* \mathcal{G}_r u$. Thus its actual squared singular values are the eigenvalues of

$$\mathcal{G}_r^{-1/2} T_r^* \mathcal{G}_{-r} T_r \mathcal{G}_r^{-1/2}. \quad (\text{N6})$$

Proof. For fixed current coefficient u , all lifts in M_r are (v, u) with lower-grade coefficient v . Their squared norms are

$$v^* \mathcal{H}_{<<} v + v^* \mathcal{H}_{<r} u + u^* \mathcal{H}_{r<} v + u^* \mathcal{H}_{rr} u.$$

Completing this exact quadratic form gives the unique minimum at $v = -\mathcal{H}_{<<}^{-1} \mathcal{H}_{<r} u$ and the value $u^* \mathcal{G}_r u$, proving (N4). The positive definiteness of $\mathcal{H}$ proves the stated inverse exists and $\mathcal{G}_r > 0$. Block elimination with these same coefficients yields $\det \mathcal{H} = \prod_r \det \mathcal{G}_r$ inductively. Since C is V_{act}^{-1} followed by a permutation, $\det \mathcal{H} = |\det V_{\text{act}}|^{-2} \det G_N$, proving (N5) with the full original root-difference determinant retained. The induced map T_r is the one already computed in (N2); writing its two quadratic forms proves (N6). No cross term was replaced by a rank count. $\square$

This gives a fully specified recipient for Deligne's nilpotent filtration within the old split-zero jet geometry. It also identifies the exact analytic matrices that a mixed-state estimate has to control. The integers in (N1) are nilpotent grades; arithmetic purity would add an assertion about eigenvalue magnitudes, which is neither a consequence of these integers nor included in this result.

7 Repair of the earlier finite-prime boundary lift

The earlier primitive-defect and split-zero-defect manuscripts use the finite-prime extension below in their semilocal arithmetic boundary construction. Their proposed invariant scalar cutoff is not by itself an element of the crossed product: the quotient unit is a rank-one Morita projection, rather than a constant function in $C_0(\mathbb{R})$. The following supplies that projection, its actual self-adjoint lift, and its entire connecting-map calculation. The sign convention remains the source's increasing logarithmic coordinate.

Fix a prime p , put $\ell = \log p$, and retain

$$Y = (\mathbb{Z} \cup \{\infty\}) \times \mathbb{R}, \quad T(m, y) = (m + 1, y + \ell), \quad T(\infty, y) = (\infty, y + \ell),$$

where $m \rightarrow +\infty$ converges to the adjoined point. Set

$$I = C_0(\mathbb{Z} \times \mathbb{R}) \rtimes_T \mathbb{Z}, \quad E = C_0(Y) \rtimes_T \mathbb{Z}, \quad Q_p = C_0(\mathbb{R}) \rtimes_{y \rightarrow y + \ell} \mathbb{Z}. \quad (\text{K1})$$

Thus $0 \rightarrow I \rightarrow E \rightarrow Q_p \rightarrow 0$ is the original extension. Use the crossed-product convention $UfU^{-1} = f \circ T^{-1}$, so that for finite coefficient families

$$(a * b)_n(z) = \sum_k a_k(z) b_{n-k}(T^{-k}z), \quad (a^*)_n(z) = \overline{a_{-n}(T^{-n}z)}. \quad (\text{K2})$$

Choose a nonnegative $c \in C_c(\mathbb{R})$ positive on $[-\ell, \ell]$ and put

$$\chi(y) = \frac{c(y)}{\sqrt{\sum_{n \in \mathbb{Z}} c(y - n\ell)^2}}.$$

The denominator is a strictly positive continuous periodic function; all sums are locally finite. Hence $\chi \in C_c(\mathbb{R})$ and $\sum_n \chi(y - n\ell)^2 = 1$ at every y . Define

$$P_n(y) = \chi(y) \chi(y - n\ell). \quad (\text{K3})$$

Only finitely many n occur, and each coefficient has compact support. Let $\phi(t) = (1 + e^t)^{-1}$ and define

$$H_n(m, y) = \phi(y - m\ell)P_n(y), \quad H_n(\infty, y) = P_n(y). \quad (\text{K4})$$

Theorem 7.1 (Actual finite-prime connecting morphism). *The family P is a full projection in Q_p whose corner is $PQ_pP \cong C(\mathbb{R}/\ell\mathbb{Z})$; it represents the unit class under the original circle Morita equivalence. The family H is an element of E , is self-adjoint with $0 \leq H \leq 1$, and lifts P . Under the explicit identification $I \cong C_0(\mathbb{R}, \mathcal{K}(\ell^2\mathbb{Z}))$, the exponential $\exp(2\pi iH)$ has Fredholm determinant*

$$\det(\exp(2\pi iH)(t)) = e^{2\pi i\phi(t)}.$$

Consequently the original connecting homomorphism, with counterclockwise winding positive, is

$$\boxed{\partial_p([P]) = -1}. \quad (\text{K5})$$

Proof. The involution formula (K2) gives $P^* = P$, and

$$(P * P)_n(y) = \chi(y)\chi(y - n\ell) \sum_k \chi(y - k\ell)^2 = P_n(y),$$

so P is a projection in the actual quotient algebra. The regular representation over an orbit $y + \ell\mathbb{Z}$ sends it to the rank-one projection onto the unit vector $v_y = (\chi(y + r\ell))_{r \in \mathbb{Z}}$. If y is increased by ℓ these vectors and the representations are related by the same bilateral shift. Compression by v_y therefore gives a scalar continuous function on $\mathbb{R}/\ell\mathbb{Z}$. Conversely every continuous periodic scalar function times P belongs to the corner, giving the claimed isomorphism. The projection is full: locally on this compact orbit space its nonzero continuous unit vector generates the rank-one operators, and finite-rank matrices generated from these are dense in the compact-operator fibers. A finite partition of unity on the circle then gives the whole algebra as the closed ideal generated by P . This is the concrete free-proper-action Morita construction of Green [13], not an identification of a constant multiplier with an algebra projection.

For each nonzero coefficient in (K4), y stays in the fixed compact support of χ . As $m \rightarrow +\infty$, $y - m\ell \rightarrow -\infty$ uniformly on that support, and $\phi(y - m\ell) \rightarrow 1$, proving continuity at the adjoined stratum. As $m \rightarrow -\infty$ it tends uniformly to zero. It follows that $H_n \in C_0(Y)$. The integer n -support is finite, so $H \in E$. The scalar $t = y - m\ell$ is invariant under T on the generic part. Consequently (K2) proves self-adjointness and

$$(H^a)_n(m, y) = \phi(t)^a P_n(y) \quad (a \geq 1),$$

with the corresponding boundary value $P_n(y)$.

For completeness, the generic algebra identification is given on the dense finite convolution algebra by

$$(\pi_t(a))_{r,s} = a_{r-s}(r, t + r\ell). \quad (\text{K6})$$

Substituting (K2) verifies multiplication and adjoints entry by entry. If a coefficient has finite m -support and compact y -support, its image is a compactly supported continuous finite matrix family. Conversely a family $f(t)e_{r,s}$ is obtained from the single coefficient $a_{r-s}(m, y) = \mathbf{1}_{\{r\}}(m)f(y - r\ell)$. These elements are dense on both sides. Thus the usual crossed-product completion gives the displayed isomorphism (equivalently the explicit free-proper-action case of [13]). Under it

$$\pi_t(H) = \phi(t) v_t v_t^*, \quad \|v_t\| = 1.$$

At every boundary orbit the representation gives the projection P . These fiber representations detect positivity; hence $0 \leq H \leq 1$.

The exponential formula is now exact, with no missing cross terms:

$$\pi_t(\exp(2\pi iH)) = I + (e^{2\pi i\phi(t)} - 1)v_t v_t^*. \quad (\text{K7})$$

The scalar coefficient tends to zero at both ends of $\mathbb{R}$, so this is a unitary in the unitization of the original ideal. Its rank-one Fredholm determinant is the asserted scalar function. That determinant represents its $K_1(C_0(\mathbb{R}) \otimes \mathcal{K})$ class: one can deform v_t to a fixed unit vector by normalized straight-line interpolation, since v_t and any fixed coordinate unit vector are nonnegative and the interpolation never vanishes. The coefficient of the rank-one projection vanishes at the two ends uniformly throughout this homotopy. This supplies an explicit homotopy to a fixed rank-one loop with the same determinant.

By the exponential definition of the boundary map, $\partial_p[P]$ is represented by this unitary. When t increases from $-\infty$ to $+\infty$, ϕ decreases from one to zero. The determinant therefore makes exactly one clockwise circuit. Its counterclockwise degree is -1 , proving (K5). $\square$

The actual visible semilocal receiving map

The earlier source specifies the visible semilocal space after quotient by the finite local unit groups. Its map to (K1) can also be given explicitly. Let $S = \{p_1, \dots, p_n, \infty\}$, put $\ell_i = \log p_i$, and let

$$X_S = \{(m_1, \dots, m_n, x) \in (\mathbb{Z} \cup \{\infty\})^n \times \mathbb{R} : \text{at most one of } m_i = \infty \text{ or } x = 0\}.$$

It carries the subspace topology from the displayed product. The group $\Gamma = \mathbb{Z}^n \times \{\pm 1\}$ acts by

$$(a, \epsilon)(m, x) = (m + a, \epsilon e^{\sum_i a_i \ell_i} x), \quad \infty + a_i = \infty. \quad (\text{K8})$$

This is precisely the action of rational S -units on the local valuations and real coordinate after finite local unit quotient. Let $A_S = C_0(X_S) \rtimes \Gamma$, let $A_\emptyset$ be the ideal of its all-nonzero stratum, and let B_S be the quotient algebra of the one-zero strata.

Theorem 7.2 (Full primitive boundary row for the specified space). *The extension $0 \rightarrow A_\emptyset \rightarrow A_S \rightarrow B_S \rightarrow 0$ has, with the original logarithmic orientation,*

$$\delta_0 : \mathbb{Z}^{n+2} \longrightarrow \mathbb{Z}, \quad \delta_0(k_1, \dots, k_n, k_+, k_-) = - \sum_i k_i - k_+ - k_-. \quad (\text{K9})$$

In particular $K_0(A_S) \cong \ker \delta_0 \cong \mathbb{Z}^{n+1}$ and $K_1(A_S) \cong \mathbb{Z}^n$. Each finite-prime component is received from (K5) by an explicit equivariant coordinate map, and the two archimedean components each have the same sign -1 .

Proof. On the invariant open edge containing the generic stratum and the p_i -zero stratum, all m_j for $j \neq i$ are finite and $x \neq 0$. Use the actual coordinates

$$z_j = m_j \ (j \neq i), \quad m = m_i, \quad y = \log |x| - \sum_{j \neq i} m_j \ell_j, \quad s = \text{sign } x. \quad (\text{K10})$$

They give a homeomorphism to $\mathbb{Z}^{n-1} \times Y_{p_i} \times \{\pm 1\}$. The inverse is $m_j = z_j$ for $j \neq i$, $m_i = m$, and $x = s \exp(y + \sum_{j \neq i} z_j \ell_j)$, proving the assertion also at $m_i = \infty$. Under (K8), the other prime generators translate only their z_j , the sign generator translates only s , and the p_i generator acts only by $(m, y) \mapsto (m + 1, y + \ell_i)$. Taking crossed products therefore gives the literal tensor factor

$$\mathcal{K}(\ell^2 \mathbb{Z}^{n-1}) \otimes E_{p_i} \otimes M_2(\mathbb{C}),$$

with matching generic ideal and boundary quotient. The compact factor identification follows directly from the matrix units for translation of a discrete group on itself. Removing these compact factors is the explicit Morita map, and the original generic logarithmic coordinate is

$$t = \log |x| - \sum_j m_j \ell_j = y - m_i \ell_i.$$

Thus the orientation is exactly the one in (K5). Naturality of the exponential boundary construction yields the component -1 .

On the invariant archimedean edge every m_i is finite and x may be zero. Use instead $u = x \exp(-\sum_i m_i \ell_i)$. This is an equivariant homeomorphism to $\mathbb{Z}^n \times \mathbb{R}$: prime generators translate the first factor only, and the sign acts by $u \mapsto -u$. Hence its extension is a compact-operator tensor factor times

$$0 \rightarrow C_0(\mathbb{R}^\times) \rtimes \{\pm 1\} \rightarrow C_0(\mathbb{R}) \rtimes \{\pm 1\} \rightarrow C^*(\{\pm 1\}) \rightarrow 0.$$

Write $p_\pm = (1 \pm U)/2$ in the quotient. Choose an even continuous real-valued $\psi(u) = e^{-u^2}$, with $\psi(0) = 1$ and $\psi(u) \rightarrow 0$ as $|u| \rightarrow \infty$. The actual self-adjoint lifts are $\psi p_\pm$ in this crossed product. On $u > 0$ the orbit representation gives the rank-one matrices $P_\pm = \frac{1}{2} \begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}$. The exponential determinant is $e^{2\pi i \psi(u)}$ for either choice. With $t = \log u$, it goes from phase 2π to zero as t goes from $-\infty$ to $+\infty$, giving -1 for both classes. This logarithmic coordinate is again exactly the common generic coordinate above.

The boundary consists of the disjoint n finite-prime strata and the archimedean stratum. It has groups $K_0(B_S) = \mathbb{Z}^{n+2}$ and $K_1(B_S) = \mathbb{Z}^n$ by the circle corners and the two sign projections just constructed. The generic ideal is stably $C_0(\mathbb{R})$, so its K_0 is zero and its K_1 is $\mathbb{Z}$. The extension and all edge inclusions have the same generic ideal; restriction to each summand of $K_0(B_S)$ therefore gives exactly the computed component. This proves (K9). Since the row is surjective, the six-term exact sequence gives $K_0(A_S) = \ker \delta_0$ and $K_1(A_S) \cong K_1(B_S)$, proving the remaining assertions. For $n = 1$ the kernel is the actual primitive A_2 lattice in $\mathbb{Z}^3$, with its three oriented generators. $\square$

Thus the actual model calculation and the visible semilocal receiving maps are both complete. This is a theorem for the specified original finite-unit-quotiented space X_S , not a claim about an unspecified crossed product. It restores the complete boundary morphism on this earlier branch of the Split Zero RH programme, while leaving arithmetic metric positivity and eigenvalue bounds to their genuine analytic calculations.

8 Denominator-history inversion reconstructs the original source

The earlier arithmetic-curve manuscript's incidence-inversion theorem is the exact finite divisor identity $\sum_{d|n} \mu(d) = \mathbf{1}_{\{n=1\}}$. Its analytic receiver in the original physical coordinate supplies a useful description of the entire source quotient, including the part not detected by a fixed finite packet. No coordinate or source measure is changed. In this section the same theta formula is also defined on the space of smooth functions on $(0, \infty)$ that are rapidly decreasing with all D -derivatives at infinity and are $O(x^{-1})$ with those derivatives at zero. Its defining series converges locally on this space; its restriction to the original V is the original Θ .

Theorem 8.1 (Exact original-source reconstruction). *For $F \in \mathcal{B}$ define*

$$(\mathcal{I}_\mu F)(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(nx) \quad (x > 0). \quad (\text{MI1})$$

The sum and all its D -derivatives converge absolutely locally uniformly on $(0, \infty)$. Its value is smooth, rapidly decreasing with all D -derivatives at infinity, and is $O(x^{-1})$ with all D -derivatives at zero. It satisfies

$$\Theta \mathcal{I}_\mu F = F, \quad \mathcal{I}_\mu \Theta \phi = \phi \quad (\phi \in V), \quad D \mathcal{I}_\mu = \mathcal{I}_\mu D. \quad (\text{MI2})$$

Consequently the original algebraic source quotient has the exact operator-compatible presentation

$$Q = \mathcal{B}/\Theta V \cong \mathcal{I}_\mu(\mathcal{B})/V. \quad (\text{MI3})$$

The obstruction on the right is membership in the original even Schwartz zero-moment space, not inversion of the theta sum on positive x . On every line $\Re w > 1$,

$$\mathcal{M}(\mathcal{I}_\mu F)(w) = \frac{\mathcal{M}F(w)}{2\zeta(w)}. \quad (\text{MI4})$$

Proof. For each compact positive x -interval and every r , the summand $D^r F(nx)$ is bounded by $C_B n^{-B}$ for every $B > 1$. This proves absolute local convergence and differentiation. For $x \geq 1$ the same bound is $C_B n^{-B} x^{-B}$, proving rapid decrease. For $g = D^r F$, both g and its ordinary derivative are integrable on $(0, \infty)$. On the interval $((n-1)x, nx)$ the fundamental theorem of calculus gives

$$|g(nx)| \leq x^{-1} \int_{(n-1)x}^{nx} |g(y)| dy + \int_{(n-1)x}^{nx} |g'(y)| dy.$$

Sum over n to obtain $|D^r \mathcal{I}_\mu F(x)| \leq (\|g\|_1/x + \|g'\|_1)/2$, proving the zero-end estimate.

For fixed $x > 0$, the double sum involved in either composition is absolutely convergent: its absolute value is bounded by $\sum_{k \geq 1} d(k) |F(kx)|$ or the analogous sum for ϕ . Use $d(k) \leq k$ and rapid decrease to verify convergence. Regrouping by $k = mn$ then gives (MI2) by the classical finite divisor identity $\sum_{d|k} \mu(d) = \mathbf{1}_{k=1}$, documented in [4, Chapter 2] and NIST DLMF (27.5.2). The derivative statement follows from the already proved locally uniform convergence. In particular $V \subset \mathcal{I}_\mu(\mathcal{B})$, and (MI2) makes the two induced maps in (MI3) inverse. This proves both injectivity and surjectivity without replacing the original V . Finally absolute integrability on $\Re w > 1$ permits substitution $y = nx$ in each Mellin integral. The absolutely convergent Euler identity $\sum_n \mu(n) n^{-w} = 1/\zeta(w)$ proves (MI4). $\square$

Proposition 8.2 (Complete primary source singularities of the actual lift). *For the original Rzu of (T2), put $\psi_u = \mathcal{I}_\mu Rzu$. Its exact equation and Mellin transform are*

$$h_Z(D)\psi_u = r_Z(u)(D)\phi_*, \quad \mathcal{M}\psi_u(w) = \frac{r_Z(u)(w)}{h_Z(w)} \mathcal{M}\phi_*(w), \quad (\text{MI5})$$

where $\mathcal{M}\phi_*(w) = \frac{1}{2}w(w-1)\pi^{-w/2}\Gamma(w/2)$. Write the proper rational function in its unique original partial fractions

$$\frac{r_Z(u)(w)}{h_Z(w)} = \sum_{\rho \in Z} \sum_{j=1}^{m_\rho} \frac{b_{\rho,j}}{(w-\rho)^j}.$$

Then, without discarding any primary or multiplicity,

$$\psi_u(x) = \sum_{\rho,j} \frac{b_{\rho,j} x^{-\rho}}{(j-1)!} \int_x^\infty \phi_*(y) y^{\rho-1} (\log(y/x))^{j-1} dy. \quad (\text{MI6})$$

Its full singular part at zero is

$$\sum_{\rho,j} \frac{b_{\rho,j} x^{-\rho}}{(j-1)!} \sum_{a=0}^{j-1} \binom{j-1}{a} (-\log x)^{j-1-a} (\mathcal{M}\phi_*)^{(a)}(\rho). \quad (\text{MI7})$$

The exact remainder obtained by subtracting (MI7) is $O(x^2)$ at zero with all D -derivatives and has a convergent even Taylor series there. Also $\int_0^\infty \psi_u(x) dx = 0$. Thus (MI6) retains the original zero-integral condition while explicitly displaying its failure of smoothness at zero for every nonzero finite class.

Proof. Apply (MI2) and (MI4) to (T2). The identity $g/(2\zeta) = \mathcal{M}\phi_*$ gives (MI5), including the factor $1/2$. Iterating the integral operator S_ρ in the proof of (T2), now without a vanishing Mellin moment at ρ , gives

$$S_\rho^j \phi_*(x) = \frac{x^{-\rho}}{(j-1)!} \int_x^\infty \phi_*(y) y^{\rho-1} (\log(y/x))^{j-1} dy.$$

This follows by Fubini over $x < y_1 < \dots < y_j$: the ordered simplex in the logarithmic variables has volume $(\log(y_j/x))^{j-1}/(j-1)!$. Its Mellin transform on $\Re w > 1$ is $\mathcal{M}\phi_*(w)/(w-\rho)^j$. The proper partial fractions therefore give (MI6) by Mellin uniqueness on that line.

The integral from zero to infinity converges for each actual $0 < \Re \rho < 1$, since $\phi_*(y) = O(y^2)$ at zero and decreases rapidly at infinity. Expanding its logarithmic power gives exactly (MI7). The remaining integral from zero to x is bounded by a constant times $x^{\Re \rho + 2}$ before multiplication by $x^{-\rho}$; substituting $y = xv$ proves the same bound for every D -derivative. More explicitly, if $\phi_*(x) = \sum_{n \geq 1} c_{2n} x^{2n}$ near zero, that remainder for $S_\rho^j \phi_*$ is precisely

$$(-1)^j \sum_{n \geq 1} \frac{c_{2n}}{(\rho + 2n)^j} x^{2n},$$

convergent near zero. The signs follow from integrating $(\log v)^{j-1}$ on $0 < v < 1$.

Formula (MI6) is integrable at zero because every actual $\Re \rho < 1$, including all logarithmic powers, and at infinity by the tail estimates. Evaluation of its Mellin transform at $w = 1$ is legitimate. Since $h_Z(1) \neq 0$ and $\mathcal{M}\phi_*(1) = 0$, its integral is zero. Finally if ψ_u were in V , (MI2) would imply $R_Z u \in \Theta V$, and its full original jets force $u = 0$ by (T2). Thus no nonzero primary class has been erased by this reconstruction. $\square$

Corollary 8.3 (Exact original dx integrability threshold). *The reconstructed finite source $\psi_u = \mathcal{I}_\mu R_Z u$ satisfies*

$$\psi_u \in L^2(\mathbb{R}_{>0}, dx) \iff u_\rho = 0 \text{ for every } \rho \in Z \text{ with } \Re \rho \geq \frac{1}{2}. \quad (\text{MI9})$$

Here $u_\rho = 0$ means the entire primary jet, not merely its value.

Proof. The local factor $\mathcal{M}\phi_*(\rho)$ is nonzero at every nontrivial zero: its explicit Gamma formula has no zero or pole there. Thus a principal part of $\mathcal{M}\psi_u = (\mathcal{M}R_Z u)/(2\zeta)$ at ρ is zero exactly when the whole jet u_ρ vanishes. Formula (MI7) shows that each remaining primary contributes a nonzero polynomial in $-\log x$ times $x^{-\rho}$, and the remainder is $O(x^2)$. If all their real parts are less than $1/2$, every term is square integrable at zero. Rapid decay at infinity proves sufficiency.

For necessity put $t = -\log x$. The singular part after multiplication by the square-root density $e^{-t/2}$ is the finite exponential polynomial $\sum_\rho e^{(\rho-1/2)t} P_\rho(t)$. Let b be the largest real part among its nonzero terms and d the largest polynomial degree at that real part. After division by $e^{(b-1/2)t} t^d$, the leading part is $A(t) = \sum_{\Re \rho = b, \deg P_\rho = d} a_\rho e^{i\Im \rho t}$, and the remaining terms tend uniformly to zero on every interval $[T, T+L]$ as $T \rightarrow \infty$ for fixed L . At least one a_ρ is nonzero. The distinct frequencies imply

$$\int_T^{T+L} |A(t)|^2 dt = L \sum |a_\rho|^2 + E(T, L), \quad |E(T, L)| \leq \sum_{\rho \neq \sigma} \frac{2|a_\rho a_\sigma|}{|\Im \rho - \Im \sigma|}.$$

Choose L so the right-hand error is at most half the positive diagonal term. This lower bound is uniform in T . The vanishing remainder preserves a fixed positive fraction of it for all large T . If $b \geq 1/2$, multiplication back by $e^{(b-1/2)t} t^d$ yields a positive lower bound on each of infinitely many disjoint such intervals, and indeed an increasing bound unless $b = 1/2, d = 0$. Their squared integrals diverge in every case. The $O(x^2)$ remainder is square integrable and cannot change that conclusion. This proves necessity without treating different frequencies as pointwise orthogonal. $\square$

There is a useful bounded tail estimate on its stated domain only. For $\sigma > 1$ in the weighted norm $\|F\|_\sigma^2 = \int_0^\infty |F(x)|^2 x^{2\sigma-1} dx$, dilation has norm $n^{-\sigma}$. The triangle inequality gives

$$\left\| \mathcal{I}_\mu F - \frac{1}{2} \sum_{n \leq L} \mu(n) F(n \cdot) \right\|_\sigma \leq \frac{L^{1-\sigma}}{2(\sigma-1)} \|F\|_\sigma. \quad (\text{MI8})$$

The original dx metric has $\sigma = 1/2$; (MI8) is not a bound in that metric. The exact maps (MI1)–(MI7), including both endpoint regions, do not require such an unsupported substitution.

9 The original theta complex over the earlier split-support space

The earlier generic-point-suspension theorem identifies the underlying space $X = \text{Spec } G(\mathbb{C})$ with two points:

$$\eta = \{\tau\}, \quad z = \{\tau, 0_{\mathbb{C}}\}, \quad \text{Open}(X) = \{\emptyset, \{\eta\}, X\}.$$

The new generic point is η and the supported closed point is z . Work here with sheaves of complex vector spaces on this underlying space; the coefficient category is specified explicitly. In particular no map making its Boolean local semiring into a complex algebra is assumed. The original infinite-dimensional spaces give algebraic sheaves, while the finite primary recipients below are constructible finite-dimensional sheaves.

Theorem 9.1 (The actual theta quotient is supported cohomology). *Define $\mathcal{F}_{\Theta}$ by*

$$\mathcal{F}_{\Theta}(X) = V, \quad \mathcal{F}_{\Theta}(\{\eta\}) = \mathcal{B}, \quad \text{res}_{X, \{\eta\}} = \Theta. \quad (\text{S1})$$

Then its local cohomology with support at z is calculated by the original complex, with its original coefficient spaces and operator:

$$R\Gamma_{\{z\}}(X, \mathcal{F}_{\Theta}) \simeq [V \xrightarrow{\Theta} \mathcal{B}] \quad \text{in degrees } 0, 1. \quad (\text{S2})$$

Thus $H_{\{z\}}^0 = 0$, $H_{\{z\}}^1 = Q$, and all higher groups vanish. For every finite actual zero packet Z , its full Mellin-jet map and canonical section give sheaf maps

$$\mathcal{F}_{\Theta} \xrightarrow{(0, J_Z)} j_! E_Z \xrightarrow{(0, R_Z)} \mathcal{F}_{\Theta}, \quad (\text{S3})$$

whose induced cohomology maps are exactly $\bar{J}_Z : Q \rightarrow E_Z$ and $\sigma_Z : E_Z \hookrightarrow Q$. Their composite on E_Z is the identity, and both cohomology maps intertwine the original D and multiplication by w .

Proof. Every open cover of X contains X itself, so a sheaf of vector spaces on X is precisely an arrow $F_z \rightarrow F_{\eta}$; morphisms are commutative squares. Its sections supported at z are the kernel of that arrow. Evaluation at either point is exact. The objects $(A \rightarrow 0)$ and $(B \xrightarrow{1} B)$ are injective: morphisms into them identify respectively with $\text{Hom}(F_z, A)$ and $\text{Hom}(F_{\eta}, B)$, exact functors of vector spaces. For a general arrow $d : A \rightarrow B$, the explicit injective resolution is

$$0 \rightarrow (A \xrightarrow{d} B) \xrightarrow{(a \mapsto (a, da), 1)} (A \oplus B \xrightarrow{\text{pr}_B} B) \xrightarrow{((a, b) \mapsto da - b, 0)} (B \rightarrow 0) \rightarrow 0. \quad (\text{S4})$$

Each component is an exact sequence, proving exactness of the sheaf sequence. The middle term is the direct sum of the two kinds of injective just identified. Applying the support-section functor gives exactly $[A \xrightarrow{d} B]$. With $d = \Theta$, this proves (S2). Its degree-zero group vanishes by the actual inverse in (MI2). The same argument for $j_! E_Z = (0 \rightarrow E_Z)$ gives its supported cohomology E_Z in degree one and zero elsewhere.

The first square in (S3) commutes because $J_Z \Theta = 0$ with all original multiplicities. The second square commutes because its closed component has zero source. Their induced maps in (S4) are the displayed original complex maps, proving the cohomology identifications and $\bar{J}_Z \sigma_Z = 1$. The first map intertwines D already on the complex. For the section, (T2) proves $DR_Z - R_Z A = \Theta \gamma_Z$, where $\gamma_Z u$ is the original scalar highest-coefficient term times ϕ_* . Thus the failure of strict intertwining is the explicitly supplied chain homotopy through the original degree-zero source V ; it vanishes on degree-one cohomology. This proves the asserted operator equivariance without changing the section. $\square$

Proposition 9.2 (The exact split semimodule underlying the same arrow). *Let $S = G(\mathbb{C})$ with additive zero τ and supported scalar zero $e = 0_{\mathbb{C}}$. On the disjoint union $M_{\Theta} = V \sqcup \mathcal{B}$, retain the ordinary vector addition within each component, and define the mixed sum by*

$$v \oplus b = b \oplus v = \Theta v + b \quad (v \in V, b \in \mathcal{B}).$$

The zero of this monoid is 0_V . A supported scalar $a \in \mathbb{C}$ acts by ordinary scalar multiplication within each component, and τ sends every element to 0_V . These operations make M_Θ an S -semimodule. Its support semilattice is

$$eM_\Theta = \{0_V, 0_{\mathcal{B}}\}, \quad 0_V < 0_{\mathcal{B}},$$

with transport map exactly Θ . The map

$$\pi : M_\Theta \longrightarrow Q \sqcup \{\tau\}, \quad \pi(v) = \tau, \quad \pi(b) = [b], \quad (\text{S5})$$

is an S -linear surjection and identifies its target with the Bourne quotient by the subsemimodule V . In particular the unsupported zero fiber is V , whereas the supported-zero fiber is the separate subset $\Theta V \subset \mathcal{B}$.

Proof. The support tag of a sum is the maximum of its two tags. Once a $\mathcal{B}$ tag occurs, the value of any finite sum is the sum in $\mathcal{B}$ of its $\mathcal{B}$ terms and of the images under Θ of its V terms. This formula proves associativity, commutativity and the stated identity. Linearity of Θ proves distributivity of supported scalars across every combination of tags. For a scalar sum involving τ , the scalar acts as the unchanged supported scalar and the other summand contributes 0_V , again the identity. For a scalar product involving τ , both sides are 0_V . The remaining scalar laws are the vector-space laws, including the case of supported scalars whose sum is the supported zero e . This verifies all semimodule axioms. Its two supported zero vectors are distinct by the disjoint-union construction, and $0_V \oplus 0_{\mathcal{B}} = 0_{\mathcal{B}}$.

Give $Q \sqcup \{\tau\}$ the usual split semimodule structure: vector addition on Q , additive identity τ , supported scalar action on Q , and τ acting by the zero map to τ . Since $q\Theta = 0$, (S5) preserves the mixed sum; it preserves all other operations directly. Surjectivity follows from that of q . In the Bourne relation, $x \sim y$ means $x \oplus v = y \oplus v'$ for some $v, v' \in V$. Both sides have the same support tag only when x, y have the same tag. All elements of V become equivalent by their additive inverses. For $b, b' \in \mathcal{B}$ the relation is precisely $b - b' \in \Theta V$. Thus its equivalence classes are one unsupported class and the supported vector quotient Q , proving the exact quotient claim and both fibers. The original D on both components preserves these operations because $D\Theta = \Theta D$, so it also descends to the original operator on Q . $\square$

The earlier support-diagram theorem therefore has a concrete receiver on the entire original theta complex, and the earlier split-spectrum theorem has a concrete supported-cohomology receiver on the same arrow. These are explicit functors into different stated coefficient categories. The sheaf (S1) is not silently obtained by quasi-coherent semiring localization: its generic complex vector space and the Boolean stalk of the structure semiring have different types. The finite maps (S3), the chain homotopy, and the split quotient (S5) specify exactly how the programme's analytic and support data meet.

10 Actual reflection, primary signs, and the unweighted Hilbert completion

The compactified-shadow manuscript supplies the full-multiplicity divisor and its Klein-four symmetry. We now realize that symmetry on the original theta complex, retaining the signs on derivative slots. The resulting calculation also determines precisely what its bare $L^2(dx)$ completion retains. The Hilbert space and the physical coordinate remain the original ones throughout.

Theorem 10.1 (Reflection on the original complex and every full primary). *On $\mathcal{B}$ define*

$$(\mathcal{I}F)(x) = x^{-1}F(1/x), \quad (\mathcal{C}F)(x) = \overline{F(x)}. \quad (\text{RC1})$$

Use the source Fourier convention $\widehat{\phi}(y) = \int_{\mathbb{R}} \phi(x)e^{-2\pi ixy} dx$. Then $\widehat{V} = V$, and

$$\mathcal{I}\Theta\phi = \Theta\widehat{\phi}, \quad \mathcal{C}\Theta\phi = \Theta\overline{\phi}, \quad D\mathcal{I} = \mathcal{I}(1 - D). \quad (\text{RC2})$$

The operator $\mathcal{I}$ is a self-adjoint unitary involution on $L^2(dx)$, while $\mathcal{C}$ is an antiunitary involution. They commute on $\mathcal{B}$, descend to Q , and preserve its finite primary subspace $\sigma_Z E_Z$ whenever Z is closed under reflection and conjugation, with every original multiplicity retained.

Write a jet as $u_\rho(t) = \sum_{j < m_\rho} u_{\rho,j} t^j$. The exact induced actions are

$$(\mathbf{W}u)_{\rho,j} = (-1)^j u_{1-\rho,j}, \quad (\mathbf{C}u)_{\rho,j} = \overline{u_{\bar{\rho},j}}. \quad (\text{RC3})$$

For the original jet map and canonical section,

$$J_Z \mathcal{I} = \mathbf{W} J_Z, \quad J_Z \mathcal{C} = \mathbf{C} J_Z, \quad \mathcal{I} R_Z = R_Z \mathbf{W}, \quad \mathcal{C} R_Z = R_Z \mathbf{C}. \quad (\text{RC4})$$

In particular, if $f_r = D^r f_0$, then

$$\mathcal{I} f_0 = f_0, \quad \mathcal{I} f_r = (1 - D)^r f_0 = \sum_{j=0}^r (-1)^j \binom{r}{j} f_j. \quad (\text{RC5})$$

Proof. Fourier transformation preserves even Schwartz functions and interchanges the conditions $\phi(0) = 0$ and $\int \phi = 0$. Poisson summation, with these two terms zero, gives (RC2). The same formula also follows from the fully specified source construction in [10, Lemma 4.1]. The chain rule gives $D\mathcal{I} = \mathcal{I}(1 - D)$ directly. Both endpoints defining $\mathcal{B}$ remain rapidly decreasing under $\mathcal{I}$ and $\mathcal{C}$. The substitution $y = 1/x$ proves $\int |x^{-1} F(1/x)|^2 dx = \int |F(y)|^2 dy$. Since $\mathcal{I}^2 = 1$, it is self-adjoint; the other Hilbert and commutation assertions follow directly from the displayed formulas.

In the original Mellin convention, $\mathcal{M}\mathcal{I}F(w) = \mathcal{M}F(1 - w)$ and $\mathcal{M}\mathcal{C}F(w) = \overline{\mathcal{M}F(\bar{w})}$. Taylor expansion at each actual root proves (RC3), including every factor $(-1)^j$, and the first two identities of (RC4). For $h = h_Z$ of degree d , root stability and multiplicities give $h(1 - w) = (-1)^d h(w)$ and $\overline{h(\bar{w})} = h(w)$. Retain $g = 2\xi$, so $g(1 - w) = g(w)$ and $\overline{g(\bar{w})} = g(w)$. The original polynomial $r_Z(u)$ is the unique polynomial of degree less than d representing $(h/g)u$ modulo h . Therefore uniqueness gives

$$r_Z(\mathbf{W}u)(w) = (-1)^d r_Z(u)(1 - w), \quad r_Z(\mathbf{C}u)(w) = \overline{r_Z(u)(\bar{w})}.$$

Insert these identities in $\mathcal{M}R_Z u = g r_Z(u)/h$. Mellin uniqueness proves the last two equalities in (RC4) as actual function identities, before passage to cohomology. Finally $\mathcal{M}f_0 = g$ and its functional equation prove $\mathcal{I}f_0 = f_0$, after which the operator relation in (RC2) proves the exact binomial formula (RC5). $\square$

Theorem 10.2 (Density of the original theta derivative family). *The original family itself satisfies*

$$\overline{\text{span}\{f_r : r \geq 0\}}^{L^2(dx)} = L^2(\mathbb{R}_{>0}, dx), \quad \overline{\Theta V}^{L^2(dx)} = L^2(\mathbb{R}_{>0}, dx). \quad (\text{RC6})$$

For every fixed original finite divisor Z , the exact attained quotient metrics

$$\begin{aligned} G_M &= (\langle f_i, f_j \rangle)_{0 \leq i, j \leq M}, \\ B_M u &= (\langle f_i, R_Z u \rangle)_{0 \leq i \leq M}, \\ H_Z^{(M)} &= R_Z^* R_Z - B_M^* G_M^{-1} B_M \end{aligned} \quad (\text{RC7})$$

obey $G_M > 0$, $H_Z^{(M)} > 0$ at each finite M , and

$$H_Z^{(M+1)} \preceq H_Z^{(M)}, \quad \lim_{M \rightarrow \infty} \|H_Z^{(M)}\| = 0. \quad (\text{RC8})$$

The inner product is $\langle F, G \rangle = \int \bar{F} G dx$. Here (RC7) uses precisely the displayed relation subspace $\text{span}(f_0, \dots, f_M)$; it does not replace another programme denominator by that subspace.

Proof. The unitary logarithmic-coordinate map is $F \mapsto e^{y/2} F(e^y)$ from $L^2(dx)$ to $L^2(dy)$. Fourier Plancherel therefore identifies the original space with $L^2(\mathbb{R}, dt/(2\pi))$ by $F \mapsto \mathcal{M}F(1/2 + it)$. This use of Fourier Plancherel is classical; see [3]. Put $g_0(t) = 2\xi(1/2 + it)$. The images of f_r are $(1/2 + it)^r g_0(t)$, retaining the original generator D .

We first prove the exponential integrability actually needed below. For $s = 1/2 + it$, the alternating-series continuation obeys

$$\eta(s) = s \int_1^\infty A(y) y^{-s-1} dy, \quad A(y) = \sum_{n \leq y} (-1)^{n-1}, \quad |A(y)| \leq 1.$$

Absolute convergence of this integral gives $|\eta(s)| \leq 2|s|$. The identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$, proved first in $\Re s > 1$ and then continued through $\Re s > 0$, gives $|\zeta(1/2 + it)| \leq 2|1/2 + it|/(\sqrt{2} - 1)$. The vertical-line consequence of Stirling's formula [2, 5.11.3] gives

$$|g_0(t)| \leq C(1 + |t|)^{11/4} e^{-\pi|t|/4}.$$

Enlarging C covers the compact interval around zero. Consequently

$$\int_{\mathbb{R}} |g_0(t)|^2 e^{2a|t|} dt < \infty \quad (0 < a < \pi/4). \quad (\text{RC9})$$

The zeros of the nonzero entire function ξ are discrete, so g_0 is nonzero almost everywhere; this uses no zero-location hypothesis.

Suppose $v \in L^2(dt/(2\pi))$ is orthogonal to all the displayed polynomial multiples. Then $k(t) = v(t)g_0(t)$ belongs to $L^1(e^{a|t|}dt)$ for each $a < \pi/4$, by Cauchy–Schwarz and (RC9), and every moment of k is zero. Its bilateral Laplace transform $K(z) = \int k(t)e^{zt}dt/(2\pi)$ is holomorphic on $|\Re z| < \pi/4$: on each smaller closed strip its derivatives are dominated using a slightly larger a in (RC9). All derivatives at zero vanish, hence the identity theorem gives $K = 0$ on the strip. In particular its Fourier transform vanishes. The uniqueness theorem for the Fourier transform of an L^1 function [3] gives $k = 0$ almost everywhere. Since $g_0 \neq 0$ almost everywhere, $v = 0$. This proves density. Each $f_r = \Theta D^r \phi_*$ belongs to ΘV , proving (RC6).

Linear independence of the f_r follows in the same Mellin representation from independence of polynomials on the real line, so $G_M > 0$. Orthogonal projection onto their finite span gives exactly the Schur minimum (RC7). If $H_Z^{(M)}$ vanished on u , then $R_Z u$ would be a linear combination of the f_r . Applying the original full jet map gives $u = 0$, since it kills every f_r and $J_Z R_Z = 1$. Thus the finite minimum is positive definite. Its monotonicity follows from inclusion of the finite relation spans. Density shows that each diagonal matrix entry tends to zero; positivity bounds each off-diagonal entry by the geometric mean of the two diagonal entries. Since the fixed divisor has finite dimension, this proves operator-norm convergence in (RC8). $\square$

Proposition 10.3 (Exact comparison of the two quotient topologies). *The inclusion of the original $\mathcal{B}$ in $L^2(dx)$ induces the map*

$$Q = \mathcal{B}/\Theta V \longrightarrow L^2(dx)/\overline{\Theta V}^{L^2(dx)} = 0. \quad (\text{RC10})$$

The infimum of the original $L^2(dx)$ norm in every coset of Q is zero. Nevertheless, for each original finite divisor Z , $\overline{J}_Z \sigma_Z = 1$ on E_Z in the original cohomology. No nonzero component of J_Z extends to a continuous linear functional on $L^2(dx)$. The original rapidly decreasing topology on $\mathcal{B}$ keeps all of these finite jets continuous.

Proof. The target and coset infimum follow from (RC6); the actual inclusion and quotient maps give (RC10). More explicitly, for a fixed $u \in E_Z$ take the minimizers

$$F_{M,u} = R_Z u - \sum_{r=0}^M (G_M^{-1} B_M u)_r f_r.$$

Their original jets equal u , their classes in Q all equal $\sigma_Z u$, and their squared norms equal $u^* H_Z^{(M)} u \rightarrow 0$. A nonzero jet functional would therefore violate continuity if extended to $L^2(dx)$. In the original $\mathcal{B}$ topology its defining integral has integrand $F(x)x^{\rho-1}(\log x)^j/j!$; a sufficiently strong rapid-decay seminorm at each endpoint bounds this integral, proving continuity there. The finite section identity was proved in (T2). $\square$

This is a calculated receiving map between the algebraic/rapid-decay cohomology and its bare unweighted Hilbert completion. It does not close the RH programme. It proves that this particular completion forgets every zero class, including critical-line classes. The finite attained metrics (RC7), their original jet maps, and the original topological quotient specify precisely what remains available for further arithmetic estimates. No uniform rate in a growing divisor or in the programme's other quotient flags follows from the fixed-divisor convergence (RC8).

11 The full Hurwitz pole train and a bilateral original-jet receiver

The earlier compact CUE/Hurwitz manuscripts compute their principal parts at 0 and 1 exactly. For a recipient in the programme's original rapidly decreasing space, all the other gamma poles must also be retained. The following construction computes them and then supplies an actual bilateral lattice recipient.

Theorem 11.1 (All negative-even residues of the original compact shifts). *Let μ be the earlier compactly supported probability measure on $[0, L]$, and retain its actual shift parameter $t \geq 0$. Write*

$$\Lambda_{\mu,t}(w) = \pi^{-w/2} \Gamma(w/2) \int \zeta(w, 1+ty) d\mu(y).$$

At every integer $k \geq 1$ its residue is

$$\text{Res}_{w=-2k} \Lambda_{\mu,t} = -\frac{2(-1)^k \pi^k}{k!(2k+1)} \int B_{2k+1}(1+ty) d\mu(y). \quad (\text{HU1})$$

For $t > 0$, the residue at -2 is strictly positive unless $\mu = \delta_0$:

$$\text{Res}_{w=-2} \Lambda_{\mu,t} = \frac{2\pi}{3} \int ty(ty + \tfrac{1}{2})(ty + 1) d\mu(y). \quad (\text{HU2})$$

Consequently removing only the 0, 1 principal parts does not produce an entire Mellin transform for a nontrivial positive compact shift. The two-endpoint formula in the earlier paper continues to hold exactly on its stated two-endpoint scope.

Proof. The classical Hurwitz value is $\zeta(-n, a) = -B_{n+1}(a)/(n+1)$; see [1, Chapter 12], documented in DLMF 25.11.14. Compactness of μ permits integration of this identity and of the holomorphic factors in a neighborhood of $-2k$. The residue of $\Gamma(w/2)$ there is $2(-1)^k/k!$, and $\pi^{-w/2} = \pi^k$ at that point. These give (HU1). The exact polynomial $B_3(1+a) = a(a+1/2)(a+1)$ is positive for $a > 0$ and zero at $a = 0$, proving (HU2) and its equality case. The 0, 1 principal parts are holomorphic at -2 , so subtracting them leaves this nonzero residue unchanged. $\square$

Theorem 11.2 (Bilateral completion in the original physical coordinate). *For $0 < a < 1$ define*

$$\begin{aligned} K_a(x) &= 2 \sum_{n \in \mathbb{Z}} e^{-\pi(n+a)^2 x^2}, \\ F_a(x) &= D(D-1)K_a(x), \\ A(w) &= w(w-1)\pi^{-w/2}\Gamma(w/2). \end{aligned} \quad (\text{HU3})$$

For each closed interval $[\epsilon, 1 - \epsilon]$ with $0 < \epsilon < 1/2$, F_a belongs to the original $\mathcal{B}$ uniformly in every defining endpoint seminorm. Its entire Mellin transform is

$$\mathcal{M}F_a(w) = A(w)(\zeta(w, a) + \zeta(w, 1 - a)) = A(w)(a^{-w} + \zeta(w, 1 + a) + \zeta(w, 1 - a)). \quad (\text{HU4})$$

Thus the two earlier shifts together with the actual missing central summand cancel every negative-even gamma pole. The exact physical inversion of this family is

$$\mathcal{I}F_a(x) = 4 \sum_{n \geq 1} \cos(2\pi na) \phi_*(nx). \quad (\text{HU5})$$

This retains the full original gaussian source ϕ_* , the original coordinate x , and every integer lattice coefficient.

Proof. For a in the stated interval, every $|n + a|$ is bounded away from zero, and the gaussian sum and all its D derivatives decrease exponentially as $x \rightarrow \infty$, uniformly in a . Poisson summation with the Fourier convention in (RC2) gives

$$K_a(x) = \frac{2}{x} \sum_{m \in \mathbb{Z}} e^{-\pi m^2/x^2} e^{2\pi i m a}. \quad (\text{HU6})$$

The term $m = 0$ is exactly $2/x$ and is killed by $D(D-1)$. All other terms, with every D derivative, decrease faster than every power at zero uniformly in a . This proves membership in the original $\mathcal{B}$ at both ends, not only a formal pole cancellation.

In $\Re w > 1$, absolute integration of the gaussian series gives

$$\mathcal{M}K_a(w) = \pi^{-w/2} \Gamma(w/2) (\zeta(w, a) + \zeta(w, 1 - a)).$$

Indeed positive terms $n + a$ for $n \geq 0$ give the first Hurwitz series and absolute values of terms with $n \leq -1$ give the second. The factor 2 in K_a cancels the $1/2$ in each gaussian Mellin integral. Two integrations by parts are justified in this half-plane at both endpoints, yielding the first equality of (HU4). Its left-hand side is entire by the just-proved endpoint estimates. The classical recurrence $\zeta(w, a) = a^{-w} + \zeta(w, 1 + a)$ proves the second equality. At negative even integers, the Bernoulli reflection identity $B_{2k+1}(1 - a) = -B_{2k+1}(a)$ gives the same cancellation algebraically; it follows also directly from the gaussian proof.

Apply $\mathcal{I}$ to (HU6). It gives $\mathcal{I}K_a = 2 + 4 \sum_{n \geq 1} \cos(2\pi na) e^{-\pi n^2 x^2}$. The identity $D(D-1)\mathcal{I} = \mathcal{I}D(D-1)$ follows from (RC2). This polynomial kills the constant and maps each gaussian to the actual $\phi_*(nx)$, proving (HU5). All termwise differentiations are absolutely locally convergent. Poisson summation and the gaussian transform used here are classical, with their Fourier convention fixed in [3]; no new convention for x or for the Mellin coordinate has been introduced. $\square$

Theorem 11.3 (Every original finite primary is attained by this family). *Let Z be any nonempty finite set of actual nontrivial zeros, with their full multiplicities, and let $J_Z : \mathcal{B} \rightarrow E_Z$ be the original jet map in (T1). For every $0 < \epsilon < 1/2$ the vectors*

$$j(a) := J_Z F_a \quad (\epsilon < a < 1 - \epsilon)$$

span all of E_Z . In the original ordered jet coordinates set

$$P_{Z, \epsilon} = \int_{\epsilon}^{1-\epsilon} j(a) j(a)^* da. \quad (\text{HU7})$$

Then $P_{Z, \epsilon} > 0$, and the explicit map

$$S_{Z, \epsilon} u = \int_{\epsilon}^{1-\epsilon} F_a j(a)^* P_{Z, \epsilon}^{-1} u da \quad : E_Z \longrightarrow \mathcal{B} \quad (\text{HU8})$$

satisfies

$$J_Z S_{Z, \epsilon} = 1, \quad J_Z (S_{Z, \epsilon} - R_Z) = 0. \quad (\text{HU9})$$

The actual original finite-sector projection on its cohomology classes is $\sigma_Z \bar{J}_Z [q S_{Z, \epsilon} u] = \sigma_Z u$. The residual difference remains in the explicitly stated $\ker J_Z$; it is not identified with ΘV .

Proof. The components of $j(a)$ are real-analytic functions on $0 < a < 1$, by (HU4) and the holomorphy of the Hurwitz parameter there. Suppose a complex linear functional λ on E_Z kills them on the stated open interval. Real-analytic uniqueness extends this identity to all $0 < a < 1$.

Near $a = 0$, the sum $\zeta(w, 1 + a) + \zeta(w, 1 - a)$ and all its finitely many required w derivatives are holomorphic and bounded in a . The singular part of $\lambda j(a)$ therefore has the exact form

$$\sum_{\rho \in Z} a^{-\rho} P_{\rho}(-\log a), \quad \deg P_{\rho} < m_{\rho}. \quad (\text{HU10})$$

The original jet convention is the coefficient of $(w - \rho)^j$, so these polynomials include their original $1/j!$ factors. For each root the map from λ 's primary coefficients to the coefficients of P_{ρ} is triangular, with nonzero diagonal multipliers $A(\rho)/j!$. They are nonzero since $0 < \Re \rho < 1$, $\rho \neq 0, 1$, and gamma has no zeros. Thus a nonzero λ gives at least one nonzero polynomial.

Such an exponential polynomial cannot stay bounded as $a \rightarrow 0$. For completeness put $y = -\log a$, choose the largest real part $\beta > 0$ among its nonzero terms, and then the largest degree d at that real part. After division by $e^{\beta y} y^d$, it equals $T(y) + o(1)$, uniformly as $y \rightarrow \infty$, where $T(y) = \sum c_{\rho} e^{i\Im \rho y}$ is a nonzero finite sum with distinct frequencies. If the original expression were bounded, then $T(y) \rightarrow 0$. On the other hand direct integration gives

$$\lim_{R \rightarrow \infty} \frac{1}{R} \int_R^{2R} |T(y)|^2 dy = \sum |c_{\rho}|^2 > 0 :$$

each cross-frequency integral divided by R tends to zero, while each diagonal integral is $|c_{\rho}|^2$. This contradicts $T(y) \rightarrow 0$. Since the remainder of $\lambda j(a)$ is bounded, the identity $\lambda j = 0$ consequently forces $\lambda = 0$. This proves the spanning assertion without discarding any primary or derivative slot.

For $u \neq 0$, the continuous nonnegative integrand $|j(a)^* u|^2$ cannot vanish on the whole interval by the spanning result. Its integral is therefore positive, proving (HU7). The uniform endpoint seminorm bounds in (HU3) show that (HU8) is a genuine integral in $\mathcal{B}$. Its continuous jet map may be passed through the integral, giving $J_Z S_{Z, \epsilon} u = P_{Z, \epsilon} P_{Z, \epsilon}^{-1} u = u$. Subtracting the already proved $J_Z R_Z = 1$ proves (HU9). Passage through the original quotient q proves the stated finite-sector cohomology formula. This calculation supplies the actual map and its exact residual space, without assigning a false equality to the two sections in the entire quotient. $\square$

The finite guard ϵ is retained: no uniform statement at $a = 0$ is asserted. The interpolation matrix (HU7) is explicitly evaluated as an integral of full Hurwitz jets, not as a rank-only allocation or an arithmetic sign estimate. It gives an additional concrete test-family receiver from the older CUE/Hurwitz construction to every original finite zero sector. Its residual cohomology class and its original dx Gram matrix remain specified objects for subsequent arithmetic work.

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Circle heat in the original arithmetic quotient metrics

The earlier circle heat construction in [1] has an exact dyadic map to the theta source of the current programme. Equations D1–D3 of the corpus receiver [2] prove the source identity and its Gram error. Here we extend that construction to the full finite relation minimum, the original additive tensor source, its complete primary polynomial and every fixed restriction or quotient of the attained metric. All coordinate maps and the finite error constants are specified below. The truncations need not span the same finite relation subspace; the comparison is proved on the full coefficient space before taking either minimum. The original polynomial and both regions of every source integral are retained throughout.

12 The source, the physical measure and exact division

Retain $x > 0$, $D = -x\partial_x$, $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$ and

$$\begin{aligned} \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, & \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), & f_j &= D^j \Theta\phi_*, \\ V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^a |D^b F(x)| < \infty \text{ for all } a \in \mathbb{R}, b \geq 0\}, \\ \mathcal{M}F(w) &= \int_0^\infty F(x)x^{w-1} dx, & g(w) &= 2\xi(w) = w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w). \end{aligned} \tag{HM1}$$

The completed-zeta convention is Apostol's convention as recorded in [4]. Poisson summation gives $\Theta V \subset \mathcal{B}$: the theta sum of an even Schwartz function has near zero the two terms $x^{-1} \int \phi - \phi(0)$ and a remainder decreasing faster than every power, and those two terms vanish on V . The same argument applies after every D derivative. Since $D(D-1)e^{-\pi x^2} = \phi_*$ and D preserves both zero conditions on V , every f_j belongs to $\mathcal{B}$. For $\Re w > 1$, absolute integration of the theta series and integration by parts give

$$\mathcal{M}f_j(w) = w^j g(w). \tag{HM2}$$

Rapid decrease at both ends extends this identity to the whole plane.

Mellin Parseval in the physical measure is

$$\|F\|_{\mathcal{H}}^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\mathcal{M}F(1/2 + iy)|^2 dy. \tag{HM3}$$

For precision, put $x = e^u$ and $\tilde{F}(u) = e^{u/2}F(e^u)$. Its $L^2(du)$ norm equals $\|F\|_{\mathcal{H}}$ and its Fourier transform with positive exponential is $(2\pi)^{-1/2}\mathcal{M}F(1/2 + iy)$. Fourier Plancherel proves (HM3), including complex functions and polarized inner products. This is the convention of [3, §1.14(iv)].

Let Z be any nonempty finite set of actual zeros of g , with their full multiplicities m_ρ , and put

$$h(w) = \prod_{\rho \in Z} (w - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \prod_{\rho \in Z} \mathbb{C}[w]/(w - \rho)^{m_\rho}. \tag{HM4}$$

The map j_h takes every divided Taylor coefficient through order $m_\rho - 1$ in the literal coordinate $w - \rho$. In particular $j_h(g/h)$ is a unit in E_h ; no root difference or Taylor coefficient of this unit is omitted.

Lemma 12.1 (Division in the original source space). *For $F \in \mathcal{B}$ with $\mathcal{M}F(\rho) = 0$, define*

$$(S_\rho F)(x) = x^{-\rho} \int_x^\infty F(t)t^{\rho-1} dt.$$

Then $S_\rho F \in \mathcal{B}$, $(D - \rho)S_\rho F = F$, and $\mathcal{M}(S_\rho F)(w) = \mathcal{M}F(w)/(w - \rho)$ with its removable value at ρ . Consequently there is a unique $a_h \in \mathcal{B}$ with $\mathcal{M}a_h = g/h$.

Proof. At infinity rapid decrease of F bounds the integral and each derivative by arbitrarily high negative powers. At zero the vanishing total integral rewrites it as $-x^{-\rho} \int_0^x F(t) t^{\rho-1} dt$, giving arbitrarily high positive powers. Differentiation gives $D(S_\rho F) = \rho S_\rho F + F$ and proves the same estimates for all derivatives. Mellin integration by parts gives the transform formula for $w \neq \rho$, and both sides are entire after removal. Starting from f_0 , divide successively by the complete multiset of roots of h . Each required Mellin zero remains until its corresponding factor is removed. The result has transform g/h . Mellin injectivity, which follows from (HM3), proves uniqueness and independence of the chosen division order. $\square$

For $u \in E_h$, let $r_h(u) \in \mathcal{P}_{d-1}$ be the unique polynomial with

$$j_h r_h(u) = j_h(h/g)u, \quad R_h u = r_h(u)(D)a_h. \quad (\text{HM5})$$

Here h/g is only used as its holomorphic germ at the specified zeros; the notation means the inverse of $j_h(g/h)$. Then $\mathcal{M}R_h u = gr_h(u)/h$ and $j_h \mathcal{M}R_h u = u$ exactly.

13 The full finite prequotient and the heat replacement

Fix $N \geq d-1$. Write $\mathcal{P}_n = \mathbb{C}[w]_{\leq n}$ and $\mathcal{P}_{-1} = 0$. The following are mutually inverse coordinate maps:

$$\begin{aligned} L_N : E_h \oplus \mathcal{P}_{N-d} &\longrightarrow \mathcal{P}_N, & (u, a) &\longmapsto r_h(u) + ha, \\ L_N^{-1} : P &\longmapsto \left(j_h((g/h)P), \frac{P - r_h(j_h((g/h)P))}{h} \right). \end{aligned} \quad (\text{HM6})$$

Indeed the first component of the inverse recovers u since $j_h g = 0$; the second is a polynomial since its numerator has all the required Taylor zeros. Both compositions are the identity, including when the relation summand is zero. Use the original jet coordinates and the ordinary power coefficients for every matrix below. Define

$$\begin{aligned} F_{N-d}a &= a(D)f_0, \quad \mathcal{R}_N = [R_h, F_{N-d}], \quad \mathcal{P}_N = [0, F_{N-d}], \quad \mathbb{G}_N = \mathcal{R}_N^* \mathcal{R}_N > 0, \\ \mathcal{M}(\mathcal{R}_N L_N^{-1}P) &= \frac{g}{h}P, \quad \left\| \mathcal{R}_N L_N^{-1}P \right\|_{\mathcal{H}}^2 = \int_{\mathbb{R}} |P(1/2 + iy)|^2 w_h(y) dy, \\ w_h(y) &= \frac{1}{2\pi} |(g/h)(1/2 + iy)|^2. \end{aligned} \quad (\text{HM7})$$

This is the original complete one-factor polynomial source. Positivity follows because the nonzero entire function g/h cannot vanish on an interval, so a zero source norm forces $P = 0$ and then $(u, a) = 0$.

On the circle $z \in \mathbb{R}/\mathbb{Z}$, keep $h_0 = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}$ and $\mathcal{N} = -(4\pi^2)^{-1} \partial_z^2$. The old construction [1] is $C(t) = \partial_z(e^{-t\mathcal{N}}h_0)(0)$. Its Fourier coefficients give $C(t) = 4 \sum_{n \text{ odd}} e^{-tn^2}$, where the sum is over all odd integers. To avoid any dependence on a formal infinite rearrangement, for every integer $J \geq 0$ put $s = 2^{J+1}$ and use the finite identity

$$\frac{1}{4} \sum_{j=0}^J C(\pi 4^j x^2) = \vartheta(x^2) - \vartheta(s^2 x^2), \quad \vartheta(v) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 v}. \quad (\text{HM8})$$

This follows by separating odd from even integers at each step. Applying $D^{j+1}(D-1)$ (with the derivative index j independent of the summation index in (HM8)) gives the exact source replacement

$$\begin{aligned} T_s F(x) &= F(sx), \quad T_s^* T_s = s^{-1} I, \\ f_j^{[J]} &= (I - T_s) f_j = \frac{1}{4} D^{j+1} (D-1) \sum_{\ell=0}^J C(\pi 4^\ell x^2), \\ \mathcal{R}_N^{[J]} &= \mathcal{R}_N - T_s \mathcal{P}_N, \quad \mathbb{G}_N^{[J]} = (\mathcal{R}_N^{[J]})^* \mathcal{R}_N^{[J]}. \end{aligned} \quad (\text{HM9})$$

All columns of $T_s \mathcal{P}_N$ lie in ΘV . Therefore the two full column maps have the same image in $\mathcal{B}/\Theta V$ and the same full jets at every actual zero. This equality of quotient classes alone makes no assertion about either finite metric; those metrics are compared next.

14 An explicit comparison before taking any minimum

When $N \geq d$, partition the full original Gram as

$$\mathbb{G}_N = \begin{pmatrix} A & C & C^* & S \end{pmatrix}, \quad H = A - CS^{-1}C^* > 0, \quad c_N^2 = 1 + \left\| H^{-1/2}CS^{-1/2} \right\|^2. \quad (\text{HM10})$$

All norms on finite coordinate spaces here are Euclidean operator norms; the positive square roots are those of the actual matrices. When $N = d - 1$ there is no relation block and define $c_N = 0$.

Theorem 14.1 (Full finite heat comparison). *For every $N \geq d - 1$ and every $J \geq 0$, let $\delta_{N,J} = 2^{-(J+1)/2}c_N$. For $\delta_{N,J} < 1$,*

$$(1 - \delta_{N,J})^2 \mathbb{G}_N \preceq \mathbb{G}_N^{[J]} \preceq (1 + \delta_{N,J})^2 \mathbb{G}_N. \quad (\text{HM11})$$

In particular the actual minima over all relation coefficients obey

$$(1 - \delta_{N,J})^2 H \preceq H^{[J]} \preceq (1 + \delta_{N,J})^2 H, \quad u^* H^{[J]} u = \min_a \|R_h u + (I - T_s)F_{N-d}a\|_{\mathcal{H}}^2. \quad (\text{HM12})$$

The matrices in (HM10) are sufficient to choose a finite J for every specified positive error tolerance; no growth assumption on c_N is used.

Proof. For the nonempty relation block, $\mathcal{P}_N^* \mathcal{P}_N = \text{diag}(0, S)$. The nonzero eigenvalues of $\mathbb{G}_N^{-1/2} \mathcal{P}_N^* \mathcal{P}_N \mathbb{G}_N^{-1/2}$ are those of the bottom principal block of $\mathbb{G}_N^{-1}$ conjugated by $S^{1/2}$. Direct block elimination gives that block as $S^{-1} + S^{-1}C^*H^{-1}CS^{-1}$. Its conjugate is $I + S^{-1/2}C^*H^{-1}CS^{-1/2}$, whose largest eigenvalue is c_N^2 . This is also obtained by completing the square

$$\|\mathcal{R}_N(u, a)\|^2 = u^* H u + (a + S^{-1}C^*u)^* S (a + S^{-1}C^*u).$$

Thus, for every coefficient vector v ,

$$\|T_s \mathcal{P}_N v\|_{\mathcal{H}} \leq s^{-1/2} c_N \|\mathcal{R}_N v\|_{\mathcal{H}}. \quad (\text{HM13})$$

The two triangle inequalities for $\mathcal{R}_N^{[J]} v = \mathcal{R}_N v - T_s \mathcal{P}_N v$ prove (HM11). The zero-relation case is an equality. To prove (HM12), use (HM11) on every vector (u, a) and take the infimum over the same coefficient a on each side. Positive definiteness ensures the minima are attained. This proves the comparison even though $F_{N-d} \mathcal{P}_{N-d}$ and $(I - T_s)F_{N-d} \mathcal{P}_{N-d}$ need not be the same subspace of $\mathcal{H}$. $\square$

For reference, the exact correction, with both cross terms retained, is

$$\mathbb{G}_N^{[J]} - \mathbb{G}_N = -\mathcal{R}_N^* T_s \mathcal{P}_N - \mathcal{P}_N^* T_s^* \mathcal{R}_N + s^{-1} \mathcal{P}_N^* \mathcal{P}_N. \quad (\text{HM14})$$

Completing a positive quadratic form and its minimum-norm lift is the classical Schur-complement calculation of [5, Appendix C.4.1]; the preceding proof specifies its matrices in this arithmetic source.

15 The exact additive tensor source and its full primary quotient

Let $k \geq 1$ and retain each separate Mellin coordinate $w_j = 1/2 + iy_j$. On the original S -power coefficients define the literal multinomial map

$$\begin{aligned} \mathcal{S}_{k,N} : \mathbb{C}[S]_{\leq N} &\longrightarrow \mathcal{P}_N^{\otimes k}, \\ S^n &\longmapsto \sum_{n_1 + \dots + n_k = n} \frac{n!}{n_1! \dots n_k!} w_1^{n_1} \dots w_k^{n_k}, \\ \mathcal{C}_{k,N} &= (L_N^{-1})^{\otimes k} \mathcal{S}_{k,N}, \\ \mathcal{A}_{k,N} &= \mathcal{R}_N^{\otimes k} \mathcal{C}_{k,N}, \quad \mathcal{A}_{k,N}^{[J]} = (\mathcal{R}_N^{[J]})^{\otimes k} \mathcal{C}_{k,N}. \end{aligned} \tag{HM15}$$

These maps target $\mathcal{H}^{\otimes k} = L^2((\mathbb{R}_{>0})^k, dx_1 \dots dx_k)$. In particular no new product, root polynomial or coefficient normalization is inserted. The multiple Mellin transform of the first map is exactly

$$\mathcal{M}^{\otimes k}(\mathcal{A}_{k,N} P)(w_1, \dots, w_k) = P(w_1 + \dots + w_k) \prod_{j=1}^k (g/h)(w_j). \tag{HM16}$$

Tensor Parseval and Tonelli, followed by the definition of convolution, therefore prove

$$\begin{aligned} \|\mathcal{A}_{k,N} P\|^2 &= \int_{\mathbb{R}^k} |P(k/2 + i \sum_j y_j)|^2 \prod_j w_h(y_j) dy_j \\ &= \int_{\mathbb{R}} |P(k/2 + iy)|^2 w_h^{*k}(y) dy. \end{aligned} \tag{HM17}$$

The entire real line is retained. All moments are finite by (HM7) and the rapid vertical decrease of Mellin transforms of $\mathcal{B}$. The form in (HM17) is the original tensor source of [6]; denote its power-basis matrix by $\mathcal{H}_{k,N} > 0$ and the corresponding heat matrix by $\mathcal{H}_{k,N}^{[J]}$.

The finite tensor quotient class is preserved as well. Indeed, expanding the difference of the tensor maps in (HM15) by replacing factors one at a time expresses it as a sum of tensors with one factor in ΘV . Consequently both maps induce the same map into the algebraic tensor quotient $\mathcal{B}^{\otimes k} / \sum_i \mathcal{B}^{\otimes(i-1)} \otimes \Theta V \otimes \mathcal{B}^{\otimes(k-i)}$. No identification with a completed tensor quotient is needed for the following finite matrix comparison.

Theorem 15.1 (The original tensor minima receive the heat error). *For $\delta = \delta_{N,J} < 1$ one has*

$$(1 - \delta)^{2k} \mathcal{H}_{k,N} \preceq \mathcal{H}_{k,N}^{[J]} \preceq (1 + \delta)^{2k} \mathcal{H}_{k,N}. \tag{HM18}$$

Let $\chi(S)$ be the original complete primary polynomial of the packet, $q = \deg \chi$, and take $N \geq \max(d-1, q-1)$. Keep the remainder map $J_{\chi,N} : \mathbb{C}[S]_{\leq N} \rightarrow E_\chi = \mathbb{C}[S]/(\chi)$ in its original coordinates. The two actual attained quotient forms satisfy

$$\begin{aligned} G_{k,N} &= (J_{\chi,N} \mathcal{H}_{k,N}^{-1} J_{\chi,N}^*)^{-1}, \quad G_{k,N}^{[J]} = (J_{\chi,N} (\mathcal{H}_{k,N}^{[J]})^{-1} J_{\chi,N}^*)^{-1}, \\ (1 - \delta)^{2k} G_{k,N} &\preceq G_{k,N}^{[J]} \preceq (1 + \delta)^{2k} G_{k,N}. \end{aligned} \tag{HM19}$$

All original multiplicities, roots and the complete relation space $\chi \mathbb{C}[S]_{\leq N-q}$ remain unchanged in this comparison.

Proof. Conjugating (HM11) by $\mathbb{G}_N^{-1/2}$ puts every eigenvalue between $(1 - \delta)^2$ and $(1 + \delta)^2$. The eigenvalues of its k -fold tensor power are the products of k such eigenvalues. Returning to the original coordinate forms gives the corresponding bounds on $(\mathbb{G}_N^{[J]})^{\otimes k}$. Pull them back through the exact map $\mathcal{C}_{k,N}$ to obtain (HM18).

For completeness, for any positive matrix M and any surjection J , the minimum of $p^* M p$ subject to $J p = v$ is $v^* (J M^{-1} J^*)^{-1} v$. Its unique minimum-norm representative is

$p_* = M^{-1}J^*(JM^{-1}J^*)^{-1}v$; every other representative is $p_* + z$ with $Jz = 0$ and $z^*Mp_* = 0$. This proves both formulas in (HM19). Taking the infimum in (HM18) over the identical affine fibre $J_{\chi,N}p = v$ gives its comparison. Its kernel is exactly the specified full polynomial ideal by Euclidean division by χ . $\square$

For the quartet used in the current programme, this theorem retains literally

$$\begin{aligned} \rho &= \tfrac{1}{2} + \delta_\rho + i\gamma, \quad 0 < \delta_\rho < \tfrac{1}{2}, \quad \gamma > 2, \quad m_0 \geq 1, \\ k &\equiv 1 \pmod{4}, \quad k \geq 5, \quad h(w) = \prod_{\sigma \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (w - \sigma)^{m_0}, \\ e &= 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \\ \chi(S) &= \prod_{a,b=0}^k [S - k/2 - (2a - k)\delta_\rho - i(2b - k)\gamma]^e. \end{aligned} \tag{HM20}$$

In particular the dyadic s is a heat truncation parameter and is not identified with k, q , a period, a root displacement or the physical coordinate $S = k/2 + iy$.

16 Restrictions, further quotients and an attained finite budget

Let M be any one of the forms just compared, and let $M^{[J]}$ be its heat counterpart in the same coefficient space. For every fixed injective frame I and fixed surjection Λ , define

$$M_I = I^*MI, \quad Q_\Lambda = (\Lambda M^{-1}\Lambda^*)^{-1}. \tag{HM21}$$

Multiplication by I preserves both inequalities in (HM18)–(HM19). Taking the infimum over $\Lambda v = u$ preserves them by the proof of (HM19). Induction gives the same constants for every finite sequence of these fixed restrictions and quotients, including the original conductor kernel and attained observation when their original coefficient maps are used. This statement fixes the maps before changing the metric; it does not substitute metric-dependent orthonormal frames for them.

If the resulting form has rank $r > 0$, each eigenvalue relative to its original form lies between $(1 - \delta)^{2k}$ and $(1 + \delta)^{2k}$. Hence its determinant in the same coordinate frame obeys

$$\left| \log \frac{\det M_X^{[J]}}{\det M_X} \right| \leq 2kr[-\log(1 - \delta)]. \tag{HM22}$$

Rank zero contributes determinant 1 and error zero. For a finite specified return $\mathcal{L} = \sum_\ell a_\ell \log \det M_{X_\ell}$, let $B = \sum_\ell |a_\ell| r_\ell$. If its cutoff family is finite, use $c_* = \max_N c_N$ over exactly those original prequotients, the same J for the whole family, and $\delta = 2^{-(J+1)/2} c_*$. Then

$$|\mathcal{L}^{[J]} - \mathcal{L}| \leq 2kB[-\log(1 - \delta)]. \tag{HM23}$$

This includes a full original cutoff window with its literal endpoint weights, once their ranks and coefficients are inserted into B . It does not discard a rank factor from an absolute determinant estimate.

Here is a finite choice that actually attains every prescribed error $\varepsilon > 0$. If $B = 0$ or $c_* = 0$, the error is zero. Otherwise put

$$\delta_* = \min \left\{ \frac{1}{4}, \frac{\varepsilon}{4kB} \right\}, \quad J = \max \{0, \lceil 2 \log_2(c_*/\delta_*) \rceil - 1\}. \tag{HM24}$$

This gives $\delta \leq \delta_* \leq 1/4$. The integral identity $-\log(1 - \delta) = \int_0^\delta (1 - t)^{-1} dt$ gives $-\log(1 - \delta) \leq 4\delta/3$, so (HM23) is at most ε (indeed at most $2\varepsilon/3$). All matrices defining c_* are the full original finite integral Grams in (HM7), with the same arithmetic unit and the two cross blocks in (HM10). Thus (HM24) proves finite convergence of these geometric heat sources to every specified original metric return. It does not assert a bound on the growth of c_N , and in particular makes no assertion that this choice of J is logarithmic in q .

17 The actual nilpotent graded metrics and their singular values

The recovered jet-ideal chain also has the following precise receiver. For the full original sum polynomial χ , write its distinct roots as κ with their original multiplicities e_κ . Its confluent matrix in the original power basis is

$$V_{(\kappa,j),n} = \binom{n}{j} \kappa^{n-j} \quad (0 \leq n < q, \ 0 \leq j < e_\kappa),$$

with zero entries when $n < j$. Let N_κ multiply by $t = S - \kappa$ on $\mathbb{C}[t]/t^{e_\kappa}$ and put $N = V^{-1}(\bigoplus N_\kappa)V$. Define the finite increasing filtration

$$M_a E_\chi = V^{-1} \bigoplus_{\kappa} \text{span}\{t^j : e_\kappa - 1 - 2j \leq a\}. \quad (\text{HM25})$$

This is the explicit single-Jordan-block construction in Deligne [7, (1.6.1), (1.6.7)]. Indeed N lowers the displayed weight by two. For $a \geq 0$, a nonzero source grade at κ occurs precisely when $a \leq e_\kappa - 1$ and $a \equiv e_\kappa - 1 \pmod{2}$, and its vector is $t^{(e_\kappa - 1 - a)/2}$. Multiplication by N^a sends it with coefficient one to $t^{(e_\kappa - 1 + a)/2}$. Thus the exact induced map $T_a : \text{Gr}_a^M \rightarrow \text{Gr}_{-a}^M$ is an isomorphism on the stated original coordinate vectors. This proves the nilpotent identities directly for this packet.

Order the columns $V^{-1}e_{\kappa,j}$ by increasing weight, with the same fixed order within ties, and call the resulting matrix C_M . For $\mathcal{B} = C_M^* G_{k,N} C_M$, let $<$ mean all weights below a . The actual metric on M_a/M_{a-1} and its heat counterpart are

$$\mathcal{G}_a = \mathcal{B}_{aa} - \mathcal{B}_{a<} \mathcal{B}_{<<}^{-1} \mathcal{B}_{<a}, \quad \mathcal{G}_a^{[J]} = \mathcal{B}_{aa}^{[J]} - \mathcal{B}_{a<}^{[J]} (\mathcal{B}_{<<}^{[J]})^{-1} \mathcal{B}_{<a}^{[J]}. \quad (\text{HM26})$$

For the first grade the second term is absent. These are precisely restrictions followed by the quotient minima in (HM21): minimizing $(v, u)^* \mathcal{B}(v, u)$ over v gives the unique lift $v = -\mathcal{B}_{<<}^{-1} \mathcal{B}_{<a} u$ and (HM26). Consequently

$$l \mathcal{G}_a \preceq \mathcal{G}_a^{[J]} \preceq u \mathcal{G}_a, \quad l = (1 - \delta)^{2k}, \quad u = (1 + \delta)^{2k}. \quad (\text{HM27})$$

In particular no cross block between different primary states or different nilpotent grades is suppressed. The frame here is a specified complex coefficient frame, without an assertion that it is an automorphism of a free split-semiring module.

Let $\lambda_{a,i} > 0$ be the ordered squared singular values of T_a in the actual two graded metrics, and use $\lambda_{a,i}^{[J]}$ for the heat metrics. Explicitly the former are the eigenvalues of $\mathcal{G}_a^{-1/2} T_a^* \mathcal{G}_{-a} T_a \mathcal{G}_a^{-1/2}$. Equation (HM27) bounds the Rayleigh quotient for every nonzero vector between l/u and u/l times its original value. Applying the finite min-max formula (minimizing the maximum over each subspace of the appropriate dimension) gives, for each ordered eigenvalue,

$$\frac{l}{u} \lambda_{a,i} \leq \lambda_{a,i}^{[J]} \leq \frac{u}{l} \lambda_{a,i}, \quad \left| \log \frac{\lambda_{a,i}^{[J]}}{\lambda_{a,i}} \right| \leq 2k \log \frac{1 + \delta}{1 - \delta}. \quad (\text{HM28})$$

This bounds the singular values of the actual induced maps, as well as the separate graded determinants. Iterated block elimination also gives

$$\det G_{k,N} = |\det V|^2 \prod_a \det \mathcal{G}_a, \quad \det G_{k,N}^{[J]} = |\det V|^2 \prod_a \det \mathcal{G}_a^{[J]}. \quad (\text{HM29})$$

Indeed $|\det C_M| = |\det V|^{-1}$, and each elimination factor in (HM26) is the actual Schur factor of the same full Gram. These constructions implement Deligne's nilpotent filtration and its metric receivers in the original sum packet. They specify the graded operators whose estimates a mixed-cohomology argument can use. No arithmetic Frobenius operator or weight-purity theorem is inferred from the integers in (HM25).

On the literal quartet sum (HM20), all primary multiplicities are e . Each nonzero grade therefore has dimension $(k+1)^2$, and for $a \geq 0$ with $a = e - 1 \pmod{2}$, $a \leq e - 1$, the same ordering of roots makes T_a the identity between those two different graded spaces. Its squared singular values are still the eigenvalues of $\mathcal{G}_a^{-1/2} \mathcal{G}_{-a} \mathcal{G}_a^{-1/2}$; the two metrics need not coincide. In particular, for the simple-quartet case $m_0 = 1$, one has $e = 1$, and the exact specialization is

$$M_{-1} = 0, \quad M_0 = E_\chi, \quad N = 0, \quad \mathcal{G}_0 = V^{-*} G_{k,N} V^{-1}, \quad T_0 = I, \quad \lambda_{0,i} = 1. \quad (\text{HM30})$$

Thus this specialization retains the entire q -dimensional arithmetic metric in its sole grade. The nilpotent comparison gives no additional separation of its distinct root states or evaluation of their observed angles. Their full original metric is carried by (HM19) and (HM30); those cross-primary calculations remain necessary.

18 Consequences for the current calculation

The source identity from the earlier paper has now been carried through the exact maps

$$\text{circle heat} \longrightarrow (I - T_s) F_{N-d} \longrightarrow \mathcal{R}_N^{[J]} \longrightarrow \mathcal{A}_{k,N}^{[J]} \longrightarrow G_{k,N}^{[J]} \longrightarrow (G_{k,N}^{[J]})_X.$$

Equations (HM6), (HM15), (HM19) and (HM21) define each coefficient map; (HM11), (HM18), (HM22) and (HM24) supply its full quantitative control. In particular the original kernel restriction in FW51–52 of [6] and its complementary observation quotient both receive this comparison. For either entire-window determinant return, one may choose (HM24) with its full B and any $\varepsilon > 0$, such as $\varepsilon = 1$. This transfers that return with its complete signs, weights and affine minima. It supplies no missing arithmetic allocation coefficient: it constructs a convergent geometric realization of the original metric in which those coefficients are to be calculated. The exact theta-class identity precedes the comparison, so closeness of metrics is never substituted for equality of the original class.

References

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- [3] Ranjan Roy, Frank W. J. Olver, Richard A. Askey, Roderick Wong and William P. Reinhardt, Chapter 1, *NIST Digital Library of Mathematical Functions*, §1.14(iv), equations 1.14.32 and 1.14.38, <https://dlmf.nist.gov/1.14>. Used for Mellin and Parseval conventions; the change to the physical measure dx is shown in (HM3).
- [4] Tom M. Apostol, Chapter 25, *NIST Digital Library of Mathematical Functions*, equation 25.4.4, <https://dlmf.nist.gov/25.4.E4>.
- [5] Stephen Boyd and Lieven Vandenbergh, *Convex Optimization*, Cambridge University Press, 2004, Appendix C.4.1, https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf. The original cross-block calculation is proved explicitly in (HM10)–(HM13).
- [6] Split-Zero research programme, *Full-window source products and the retained class: finite quotient comparisons in the original arithmetic metric*, 19 September 2026, FW1–52, particularly FW1–4 and FW37–40, FW51–52. GitHub edition 3c44de632b2c5ead13b23619d646803d2bbffc19.

- [7] Pierre Deligne, *La conjecture de Weil: II*, Publications Mathématiques de l’IHÉS 52 (1980), 137–252, §1.6, especially (1.6.1) and (1.6.7), https://www.numdam.org/item/PMIHES_1980__52__137_0/. The complete D032 typed source accompanies this proof. Equations (HM25)–(HM29) specify and prove its finite packet and metric receiver.

Required heat depth in a fixed original packet

This working continuation of HM1–30 keeps the original fixed divisor h , its full jet space E_h , canonical section R_h and physical Hilbert space $L^2(\mathbb{R}_{>0}, dx)$. It determines the dependence of the proved heat comparison on the actual finite metric. The underlying source-density calculation was recovered as RC6–10 in `EXACT_CORPUS_RECEIVERS.tex`; its needed proof is reproduced here. The accepted HM source is preserved at GitHub commit `88f967d3c38750c57264fc5ebbf10bb463717e4e`.

19 Density in the stated physical norm

Put $D = -x\partial_x$, $f_j = D^j\Theta\phi_*$ and $g_0(t) = 2\xi(1/2+it)$, with the original ϕ_* of HM1. Mellin Parseval sends f_j to $(1/2+it)^j g_0(t)$ in $L^2(\mathbb{R}, dt/(2\pi))$. We first establish the exact density needed for the finite-metric conclusion.

The bounded alternating partial sum $A(y) = \sum_{n \leq y} (-1)^{n-1}$ gives $\eta(s) = s \int_1^\infty A(y) y^{-s-1} dy$ for $\Re s > 0$. On $s = 1/2+it$ this has absolute value at most $2|s|$. Continuation of $\eta(s) = (1-2^{1-s})\zeta(s)$ from $\Re s > 1$ then gives $|\zeta(s)| \leq 2|s|/(\sqrt{2}-1)$ on this line. Stirling's formula in its fixed vertical strip, with the full Gamma factor in 2ξ , yields

$$|g_0(t)| \leq C(1+|t|)^{11/4} e^{-\pi|t|/4}, \quad \int_{\mathbb{R}} |g_0(t)|^2 e^{2a|t|} dt < \infty \quad (0 < a < \pi/4). \quad (\text{HD1})$$

The Gamma asymptotic is the classical Stirling formula cited in [1]; the partial-summation bound was just proved.

If v is orthogonal to all the polynomial multiples of g_0 , put $b(t) = \overline{v(t)}g_0(t)$. Cauchy–Schwarz and (HD1) give $b \in L^1(e^{a|t|}dt)$ for every $a < \pi/4$, and all its polynomial moments vanish. Its bilateral transform $B(z) = \int b(t)e^{zt}dt/(2\pi)$ is holomorphic on $|\Re z| < \pi/4$: on every smaller closed strip all derivatives are dominated using a slightly larger a in (HD1). Its Taylor coefficients at zero vanish, so the identity theorem gives $B = 0$ on this strip. Fourier uniqueness for L^1 functions [2] implies $b = 0$ almost everywhere. The zeros of the nonzero entire function ξ are discrete, so $g_0 \neq 0$ almost everywhere and $v = 0$. We have proved

$$\overline{\text{span}\{f_j : j \geq 0\}}^{L^2(dx)} = L^2(dx). \quad (\text{HD2})$$

This proof uses no assumption about the real parts of the zeros.

20 The exact finite constant and the required heat scale

Fix the complete nonempty divisor h throughout this section and write $d = \deg h$. For $N \geq d$, retain the matrices of HM10:

$$\mathbb{G}_N = \begin{pmatrix} A & C_N \\ C_N^* & S_N \end{pmatrix}, \quad A = R_h^* R_h > 0, \quad H_N = A - C_N S_N^{-1} C_N^* > 0.$$

The relation columns are exactly $f_0, \dots, f_{N-d}$; the canonical section and A are independent of N . In particular every statement below fixes the divisor rather than replacing it by a growing one.

Theorem 20.1. *The full comparison constant in HM10 is exactly*

$$c_N^2 = \lambda_{\max}(H_N^{-1/2} A H_N^{-1/2}) = \sup_{u \neq 0} \frac{\|R_h u\|_{L^2(dx)}^2}{u^* H_N u}. \quad (\text{HD3})$$

Moreover H_N decreases in Loewner order to zero in operator norm, and c_N is nondecreasing for $N \geq d$ with $c_N \rightarrow \infty$. For the literal heat replacement in HM9, set $\delta = 2^{-(J+1)/2} c_N$. Then

$$\lambda_{\max}(\mathbb{G}_N^{-1/2} \mathbb{G}_N^{[J]} \mathbb{G}_N^{-1/2}) \geq (\delta - 1)_+^2. \quad (\text{HD4})$$

Thus any fixed finite upper comparison constant $B > 0$ for these full prequotients requires

$$J + 1 \geq 2 \log_2 c_N - 2 \log_2(1 + \sqrt{B}). \quad (\text{HD5})$$

The sufficient depth of HM24 and this necessary depth therefore have the same leading dependence $2 \log_2 c_N$, with their stated fixed tolerance constants retained. No rate for c_N in N is asserted.

Proof. Since $C_N S_N^{-1} C_N^* = A - H_N$, the definition in HM10 becomes

$$c_N^2 = 1 + \lambda_{\max}(H_N^{-1/2}(A - H_N)H_N^{-1/2}) = \lambda_{\max}(H_N^{-1/2} A H_N^{-1/2}).$$

Substituting $v = H_N^{1/2} u$ in its Rayleigh quotient proves the last formula in (HD3), without changing the original coefficient maps.

The space of relation columns increases with N , so its attained distance from H_N decreases. The density (HD2) makes the distance from every column of R_h to this relation space tend to zero. Hence every diagonal entry of H_N tends to zero. Positivity bounds the absolute value of every off-diagonal entry by the geometric mean of the corresponding diagonal entries. The fixed dimension is finite, so $\|H_N\| \rightarrow 0$. At each finite N it is positive definite: if $R_h u$ were in the finite relation span, the full original Mellin jet map would give $u = 0$, since it kills all the f_j . Formula (HD3) proves monotonicity of c_N . It also gives $c_N^2 \geq \lambda_{\min}(A) / \|H_N\| \rightarrow \infty$.

Let $\mathcal{P}_N = [0, F_{N-d}]$ and $s = 2^{J+1}$, as in HM7–9. The operator $U_N = \mathcal{R}_N \mathbb{G}_N^{-1/2}$ is an isometric embedding, so $\|U_N\| = 1$. By the exact computation in HM13 and $T_s^* T_s = s^{-1} I$,

$$\|T_s \mathcal{P}_N \mathbb{G}_N^{-1/2}\| = s^{-1/2} c_N = \delta.$$

The reverse triangle inequality implies $\|\mathcal{R}_N^{[J]} \mathbb{G}_N^{-1/2}\| \geq (\delta - 1)_+$. Squaring gives (HD4). If the left side of (HD4) is at most B , then $\delta \leq 1 + \sqrt{B}$, which is exactly (HD5). At fixed J the lower bound in (HD4) tends to infinity. Every map here is the full original prequotient map with the same section and coefficient fibre as HM6. $\square$

21 The receiving quotient is still taken through its original map

The lower bound (HD4) concerns the full form before minimization. Its exact receiving map is the already proved $M \mapsto (\Lambda M^{-1} \Lambda^*)^{-1}$. A lower bound on the largest eigenvalue of a full form need not survive that map. For an explicit verification, let $M_t = \text{diag}(t, 1)$ on $\mathbb{C}^2$, $M_0 = I$, and $\Lambda(z_1, z_2) = z_2$. Then $\lambda_{\max}(M_t) = t$ for $t \geq 1$, but $\Lambda M_t^{-1} \Lambda^* = 1$ and the attained quotient form is identically 1. The fixed-divisor limit above therefore proves the stated need to increase heat depth for the full-prequotient comparison; it does not assign a relative-rate failure to the re-minimized packet metrics. Those continue to obey the explicit two-sided bounds of HM19 whenever the depth in HM24 is used.

The exact representatives $R_h u - F_{N-d} S_N^{-1} C_N^* u$ keep the original full jets and theta cohomology class while their squared norms are $u^* H_N u \rightarrow 0$. Thus the inclusion of the original rapid-decay source space into $L^2(dx)$ induces the zero map to the Hausdorff quotient by the closed theta image. Its finite jet sections still split the original algebraic cohomology. These explicit maps preserve the distinction between the chosen source topology and its Hilbert completion, and leave the original growing-packet arithmetic estimates intact.

References

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Original complex response and projected-current errors

We carry HM19 into the original responses of ORE8 and the three actual centered pairings of ORE11–12. The source classes, observation map, period, arithmetic coefficient and physical coordinate remain fixed. Only the metric in their explicit evaluation is replaced by its stated heat approximation. In particular the approximating boundary columns below are the *original* columns, not newly chosen monic polynomials for the heat source.

22 Every observation cross term receives a finite bound

At an original cutoff N write $G = G_{k,N}$ and $G' = G_{k,N}^{[J]}$. Retain $\Lambda : E \rightarrow B = \text{im } \Lambda$, $K = \ker \Lambda$ and the same original two boundary vectors $a_1 = b_N$, $a_2 = b_{N+1}$ and terminal class v . HM19 proves

$$\alpha G \preceq G' \preceq \beta G, \quad \alpha = (1 - \delta)^{2k}, \quad \beta = (1 + \delta)^{2k}, \quad \delta = 2^{-(J+1)/2} c_N < 1. \quad (\text{HR1})$$

The full prequotient constant c_N and all its arithmetic cross blocks are those of HM10. Put $\eta = \max(1 - \alpha, \beta - 1) = \beta - 1$; the equality follows from the even terms in the binomial expansion of $(1 + \delta)^{2k} + (1 - \delta)^{2k} \geq 2$. Use the exact original quotient and kernel forms

$$Q = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad \Omega = \Lambda^* Q \Lambda, \quad L = G - \Omega, \quad (\text{HR2})$$

and define Q', Ω', L' with G' and the same Λ . These formulas include zero-dimensional image or kernel with the usual empty-matrix conventions. The exact minimum-norm lift proves $0 \preceq \Omega \preceq G$ and $L = G P_K$, where P_K is the G -orthogonal projection onto the original kernel. Indeed the lift of $b \in B$ is $G^{-1} \Lambda^* Q b$, whose difference from any other lift is orthogonal to it. This is the original Schur minimum, also described by Boyd and Vandenberghe [1, Appendix C.4.1].

Theorem 22.1 (Full cross-term error). *For all original coefficient vectors $x, y \in E$,*

$$\begin{aligned} |x^*(G' - G)y| &\leq \eta \sqrt{x^* G x} \sqrt{y^* G y}, \\ |x^*(\Omega' - \Omega)y| &\leq \eta \sqrt{x^* G x} \sqrt{y^* G y}, \\ |x^*(L' - L)y| &\leq 2\eta \sqrt{x^* G x} \sqrt{y^* G y}. \end{aligned} \quad (\text{HR3})$$

The observed fraction of the retained norm has the sharper ratio bound

$$\frac{\alpha}{\beta}(1 - \theta) \leq 1 - \theta' \leq \frac{\beta}{\alpha}(1 - \theta), \quad \theta = \frac{v^* L v}{v^* G v}, \quad \theta' = \frac{v^* L' v}{v^* G' v}. \quad (\text{HR4})$$

If $\Lambda v = 0$, both observed fractions are exactly zero.

Proof. Conjugating (HR1) by $G^{-1/2}$ bounds the Hermitian operator norm of $G^{-1/2}(G' - G)G^{-1/2}$ by η . Cauchy–Schwarz gives the first inequality. Taking the minimum over each unchanged fibre $\Lambda x = b$ in (HR1) gives $\alpha Q \preceq Q' \preceq \beta Q$. Pulling back gives $-\eta \Omega \preceq \Omega' - \Omega \preceq \eta \Omega$. As $0 \preceq \Omega \preceq G$, the same conjugation proves the second inequality. Subtract the two corrections in (HR2), retaining both, to obtain the third. For (HR4) use $v^* \Omega v = (v^* G v)(1 - \theta)$ and its primed identity. The numerator and denominator each have factors between α and β , giving the stated ratios. Their numerator is zero in both metrics exactly when $\Lambda v = 0$. $\square$

23 The original response ratios and their amplification

Write $E = v^* G v$, $d_i = a_i^* G a_i$ and retain the original quantities

$$F_i = a_i^* G v, \quad K_i = a_i^* L v, \quad p_i = \frac{|F_i|^2}{d_i E}, \quad U = \frac{K_1}{F_1}, \quad V = \frac{K_2}{F_2}. \quad (\text{HR5})$$

On the original terminal-class domain the two F_i are nonzero by the full source calculation RBE12–19 and FW4–5, so $0 < p_i \leq 1$. All these numbers use the full original minimum. Define F'_i, K'_i and U', V' by the same vectors and (HR2), now evaluated in G' . For an explicit finite guard choose the heat depth so $\eta < \min(\sqrt{p_1}, \sqrt{p_2})$. Such a depth always exists by HM24, with all original constants retained. Then

$$\begin{aligned} |F'_i| &\geq (\sqrt{p_i} - \eta)\sqrt{d_i E} > 0, \\ |U' - U| &\leq e_U := \frac{\eta(2 + |U|)}{\sqrt{p_1} - \eta}, \quad |V' - V| \leq e_V := \frac{\eta(2 + |V|)}{\sqrt{p_2} - \eta}. \end{aligned} \quad (\text{HR6})$$

Indeed (HR3) gives $|F'_i - F_i| \leq \eta\sqrt{d_i E}$ and $|K'_i - K_i| \leq 2\eta\sqrt{d_i E}$. The exact identity

$$\frac{K'_i}{F'_i} - \frac{K_i}{F_i} = \frac{(K'_i - K_i) - (K_i/F_i)(F'_i - F_i)}{F'_i}$$

proves every factor and denominator in (HR6).

There is also a bound using only original source energies and the observed fraction. Positivity of L , $a_i^* L a_i \leq d_i$, and $v^* L v = \theta E$ give

$$|U| \leq \sqrt{\theta/p_1}, \quad |V| \leq \sqrt{\theta/p_2}. \quad (\text{HR7})$$

Thus the right sides in (HR6) can be bounded with the original p_i, θ . Whenever the sharper original conductor bounds FW50 are used, they can replace (HR7) in exactly the same formula; no denominator or amplification is removed. For example, the finite guard $\eta \leq \frac{1}{2} \min(\sqrt{p_1}, \sqrt{p_2})$ gives the coarser completely explicit bounds $e_U \leq 6\eta/p_1$, $e_V \leq 6\eta/p_2$, since $\theta \leq 1$ and $p_i \leq 1$.

24 The three original centered pairings

Retain the original complex arithmetic coefficient $\mathfrak{c} = \mathfrak{c}_{N,\lambda}$ from ORE12. This is fixed in the following approximation of the *original* evaluation; it is not recomputed from a different source action. Put

$$\begin{aligned} P_K &= \bar{U}V, \quad P_B = (1 - \bar{U})(1 - V), \quad P_\times = 1 - P_K - P_B, \\ \Phi_X &= 2E \operatorname{Re}(\mathfrak{c}P_X), \quad \Phi'_X = 2E' \operatorname{Re}(\mathfrak{c}P'_X), \quad E' = v^* G' v, \end{aligned} \quad (\text{HR8})$$

where each primed product uses U', V' and the same displayed complex conjugations. These are the original kernel, observation and complementary cross products. Expanding each product without discarding its quadratic correction proves

$$\begin{aligned} |P'_K - P_K| &\leq b_K := |V|e_U + |U|e_V + e_U e_V, \\ |P'_B - P_B| &\leq b_B := |1 - V|e_U + |1 - U|e_V + e_U e_V, \\ |P'_\times - P_\times| &\leq b_\times := b_K + b_B. \end{aligned} \quad (\text{HR9})$$

Consequently the actual current evaluation has the explicit error

$$|\Phi'_X - \Phi_X| \leq 2E|\mathfrak{c}|[\eta|P_X| + \beta b_X], \quad X \in \{K, B, \times\}. \quad (\text{HR10})$$

To check (HR10), write its difference as $2(E' - E) \operatorname{Re}(\mathfrak{c}P_X) + 2E' \operatorname{Re}(\mathfrak{c}(P'_X - P_X))$. Use $|E' - E| \leq \eta E$ and $E' \leq \beta E$ from (HR1). The identities $\sum_X P_X = \sum_X P'_X = 1$ are exact. In particular

$$\sum_X \Phi_X = 2E \operatorname{Re} \mathfrak{c}, \quad \sum_X \Phi'_X = 2E' \operatorname{Re} \mathfrak{c}, \quad \left| \sum_X (\Phi'_X - \Phi_X) \right| \leq 2\eta E |\operatorname{Re} \mathfrak{c}|. \quad (\text{HR11})$$

For every prescribed finite cutoff family these estimates also sum with the original absolute weights, by the triangle inequality. All terms on the right are original coefficients or their proved bounds, so the errors tend to zero as $J \rightarrow \infty$ while this finite family is fixed. The small denominators in (HR6) remain visible. These formulas provide the required error control for an evaluation of the original projected signs; they do not assign those signs without evaluating the original responses.

References

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